

Why (we still need) linear regression

a deep-dive for applied and data scientists

为什么（依然）需要线性回归 — 写给应用与数据科学家的深度教程

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摘要

Linear regression is the first model everyone learns and the one almost everyone underestimates. This tutorial is not about running `lm()` in R or tuning `LinearRegression()` in scikit-learn — it answers *why* the method works: why squared error, why the normal equations, why the correlation coefficient appears inside a regression coefficient, why a straight line can measure *correlation* yet cannot assert *causation*, and why, more than two hundred years later, the d -dimensional hyperplane formula $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is still the most honest starting point of data analysis. The only prerequisites are high-school calculus and basic linear algebra; every claim is derived from scratch, and all historical references are verifiable.

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Preface

Who this book is for

This tutorial is written for **applied scientists and data scientists** with some mathematical background — you have taken linear algebra and calculus, you use `lm()`, `LinearRegression()`, or `np.linalg.lstsq` on a daily basis, and yet you feel a thin pane of glass between “knowing how to use” and “understanding”: why is the slope exactly the correlation coefficient? Why squared loss rather than absolute value? How does a model with zero training error in the $d > n$ regime manage to generalize at all? What exactly is the relationship between kernel methods and my linear regression?

If these questions sound familiar, this book is for you. It does not teach you how to call packages (that is what documentation is for); it derives every formula from scratch and hands you the **assumption list** behind every theorem — because in statistics the assumptions are the substance, and the formulas are merely their shadow. The only prerequisites are undergraduate calculus and linear algebra plus basic probability; every claim can be verified by hand, and complete solutions to all exercises are collected in the appendix.

Origins and influences

This book was written **on a whim** — one evening, the author decided to write down, in one sitting, the “why do we even learn linear regression” argument that had been circulating in his head for years, and ended up going chapter by chapter all the way to Chapter 8. But the whim stands on the accumulated influence of a body of work read over a long time. The seven scholars who influenced it most are: **Francis Bach** (Inria), whose textbook *Learning Theory from First Principles* (2024)[23] shares this book’s temperament of earning every result from first principles — the kernel-methods perspective of Chapter 8 and the optimization–statistics interlocking of Chapters 4–5 both bear its mark; **Ryan Tibshirani** (Stanford University), who turned generalized lasso, trend filtering, and conformal prediction into methodology that is both computable and auditable — precisely the spirit of the estimate-to-inference route in Chapter 6; **Jingfeng Wu** (UC Berkeley, Simons Institute), whose work showing that “gradient descent dominates ridge regression”[28] is the statistical frontier of the iteration-as-regularization story in Chapters 4–5.5; **Arthur Jacot** (NYU Courant Institute), whose neural tangent kernel[24] welded “linear regression → kernel regression → deep networks” into a single road — the content of Section 8.5; **Jianqing Fan** (Princeton University), whose high-dimensional statistical framework[27] — sparse estimation, variable screening, large covariance matrices — is the source of the from-estimation-to-inference roadmap of Chapter 6; **David Donoho** (Stanford University), founder of compressed sensing, who proved that ℓ_1 minimization exploits sparsity optimally and that underdetermined systems become solvable under sparsity assumptions[29] — Section 6.4’s “sparsity is a computable assumption” is the statistical expression of exactly this idea; **Emmanuel Candès** (Stanford University), who co-created the exact-recovery theory of compressed

sensing with Donoho and Tao, and who, through the knockoffs framework, gave complex-model variable selection reproducible false discovery rate (FDR) control[30] — the modern next station after Section 6.6’s “from point estimates to testable inference.”

How it was written, and an honest disclaimer

This book was written with substantial AI assistance. Specifically, the text was produced by Moonshot AI’s Kimi K2.8 model running in the **Kimi Work desktop** agent mode — it wrote the LaTeX source directly, compiled the PDF, rendered and inspected pages, and proofread chapter by chapter; the human expert (Haoyi Xiong) supplied the outline, the chapter content, and his own thinking — in particular his multi-year understanding of linear regression and the specific points he had personally worked on (such as the exact solution of gradient descent in the underdetermined case, the kernel-methods view under latent projections, and the precision-matrix–conditional-independence connection). One sentence of division of labor: *the human decides “what to ask” and “what to believe”; the machine does the “writing” and the “checking.”*

Readers should therefore keep three things in mind:

- (a) **The field of linear regression is vast**, and this book does not and cannot cover it all: generalized linear models, robust regression, mixed-effects models, time series, the Bayesian route, online learning, federated settings — each deserves its own deep-dive. This book covers only the main line that the authors consider impossible to skip honestly.
- (b) **The theory here does not aim for professional rigor.** Every formula is correct and derivable, but the proofs are plainly not textbook-complete (regularity conditions for asymptotic results are uniformly omitted). *Readers who work on theory should skip this book*; beginners or those who have never seen the material systematically may read it as a popularization with derivations.
- (c) **The historical and bibliographic sections were checked with care** (principal figures, years, and sources were verified online and marked in the bibliography), but if any error remains, the responsibility lies with the human authors, and corrections are welcome.

Fit the straight line first, audit the assumptions, disclose the weaknesses honestly — this book applies its own recipe to itself.

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1 Introduction: Why the Straight Line Is Still the Honest Starting Point, Two Hundred Years On

1.1 The questions this tutorial answers

Every machine learning course and every statistics textbook teaches linear regression in the first two weeks, and almost all of them teach it the same way: you are handed a formula

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y},$$

told to memorize it, and then rushed off toward “more exciting” models.

This tutorial does the opposite: every step is earned from scratch. By the end, you should be able to answer, from first principles:

- (1) Why is the slope of one-dimensional regression $\frac{\text{Cov}(x, y)}{\text{Var}(x)}$? What exactly is its relationship to the Pearson correlation coefficient? (Chapter 2)
- (2) Why can a straight line measure the *correlation* between x and y , yet tell you nothing about the *direction* of causation? (Chapter 2)
- (3) Why squared error? What exactly would be lost with a different loss? (Section 3.5)
- (4) In high dimensions, when the sample size n far exceeds the dimension d , why does least squares admit the closed-form solution $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$? When is this solution unique, and when is it optimal? (Chapter 3)
- (5) When n no longer dwarfs d , what breaks first, and what repairs it? (Section 3.6)
- (6) Putting the closed form and the gradient-descent iteration side by side, what is the exact solution of $\hat{\boldsymbol{\beta}}_t$? What does the fixed point mean? (Chapter 4)
- (7) When $d > n$ and the solutions form a manifold, where does the iteration converge? How is it related to the pseudoinverse solution? (Chapter 5)
- (8) How are regression coefficients related to the covariance and precision matrices? How does “estimation” graduate into “inference” — from estimate to inference? (Chapter 6)
- (9) When the observed features are in fact nonlinear projections of latent variables, why does linear regression behave as a kernel method? (Chapter 8)

1.2 A two-hundred-year-old idea that refuses to retire

Least squares was first published by Legendre (1805); Gauss independently proposed it and gave a full probabilistic derivation — having in fact used the method to determine the orbit of Ceres from noisy astronomical observations as early as 1801–02, and publishing his derivation only in 1809[1, 2]. The method has outlived every paradigm shift since: the computer, the bootstrap, the deep learning revolution. For exactly one reason:

The thesis of this tutorial

Linear regression is popular not because the world is linear — it rarely is. It is popular because:

- (i) a local linear approximation is often good enough (Taylor’s theorem guarantees that every smooth function is approximately linear locally);
- (ii) it is the *only* baseline that is exactly solvable, interpretable, and auditable — every other model measures itself against it;
- (iii) it is the foundation of logistic regression, generalized linear models, kernel methods, and even the output layer of neural networks.

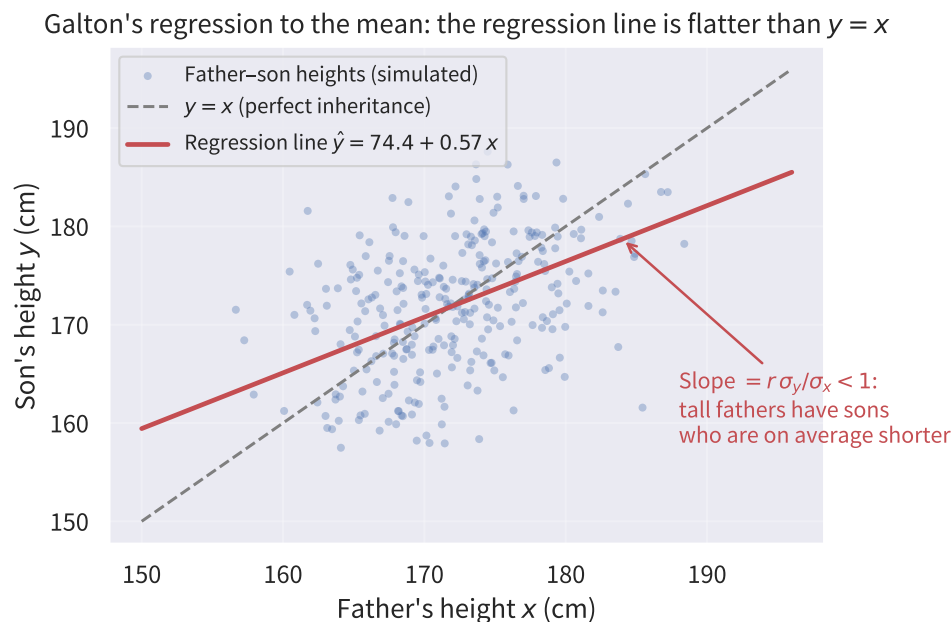


图 1: Galton’s “regression toward the mean” (recreated with simulated data): the regression line (red) is always flatter than $y = x$ (gray dashed) — the slope is $r\sigma_y/\sigma_x$, so as long as $|r| < 1$, extremes necessarily fall back. Two hundred years on, this straight line is still the default starting point of data analysis.

1.3 Prerequisites

- **Calculus:** partial derivatives, the chain rule, and the fact that extrema of smooth functions occur where derivatives vanish;
- **Linear algebra:** vectors, matrices, matrix multiplication, transpose, inverse; the reading of “matrix times vector as a linear combination of columns”;
- **Probability** (used from Chapter 2 onward): expectation, variance, covariance, correlation.

Everything else is derived in the text.

1.4 Notation, stated once

One-dimensional case: n paired observations $(x_i, y_i) \in \mathbb{R}^2$, $i = 1, \dots, n$. High-dimensional case: responses $y_i \in \mathbb{R}$, features $\mathbf{x}_i = (x_{i1}, \dots, x_{id})^\top \in \mathbb{R}^d$, design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ (rows equal to $\mathbf{x}_i^\top$), parameters $\boldsymbol{\beta} \in \mathbb{R}^d$. The error terms ε_i absorb everything a hyperplane cannot explain: measurement noise, omitted variables, genuine randomness, and the original sin of approximating a nonlinear world with a hyperplane.

Conventions:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \text{Var}_n(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad \text{Cov}_n(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}),$$

i.e., sample variance and covariance with denominator n (using $n - 1$ would only introduce an immaterial factor into every ratio). Throughout, Var and Cov without a subscript denote population quantities taken with respect to expectation, and the subscript n denotes sample quantities.

2 One-Dimensional Linear Regression: Correlation, Direction, and Causation Inside a Single Line

Start from the simplest model — no intercept, a single feature:

$$y_i = b x_i + \varepsilon_i, \quad i = 1, \dots, n, \quad (1)$$

where $b \in \mathbb{R}$ is the parameter to estimate. This chapter does two things: first, *from the regression viewpoint*, it shows how b measures the correlation between x and y ; then, *from the variance viewpoint*, it shows how the same straight line exposes the *directionality* of correlation, and why it cannot tell you causation.

2.1 Least squares: generalizing the mean into a line

Forget statistics for a moment. Write the fitting problem as an optimization problem: find b that minimizes the sum of squared residuals

$$S(b) = \sum_{i=1}^n (y_i - b x_i)^2. \quad (2)$$

Differentiate with respect to b and set to zero:

$$\frac{dS}{db} = -2 \sum_{i=1}^n x_i (y_i - b x_i) = 0 \quad \implies \quad \boxed{\hat{b} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}}$$

The second derivative $\frac{d^2 S}{db^2} = 2 \sum_i x_i^2 > 0$ (as long as x is not identically zero), so this is the unique global minimizer. No initial guess, no learning rate, no local minima — convexity settles everything at once.

Look at two extremes to build intuition. If $x_i \equiv 1$, then $\hat{b} = \bar{y}$: **least squares is the generalization of the mean** — the sample mean is precisely the constant that minimizes the sum of squared deviations. If the model has an intercept, $y_i = a + b x_i + \varepsilon_i$, exactly the same differentiation gives

$$\hat{b} = \frac{\text{Cov}_n(x, y)}{\text{Var}_n(x)}, \quad \hat{a} = \bar{y} - \hat{b} \bar{x}, \quad (3)$$

i.e., the regression line must pass through the data centroid $(\bar{x}, \bar{y})$. Read (3) slowly: *the slope is the degree to which x and y vary together, divided by the degree to which x varies on its own*. The most quoted formula in all of applied statistics is nothing more than one differentiation. Note that no error distribution has been assumed so far — least squares, up to this point, is purely geometric optimization.

2.2 The regression viewpoint: the slope *is* the correlation coefficient

Now standardize the data. Write $\sigma_x = \sqrt{\text{Var}_n(x)}$, $\sigma_y = \sqrt{\text{Var}_n(y)}$; the Pearson (1896) correlation coefficient[5] is

$$r = \frac{\text{Cov}_n(x, y)}{\sigma_x \sigma_y} \in [-1, 1]. \quad (4)$$

(By Cauchy–Schwarz, $|\text{Cov}_n(x, y)| \leq \sigma_x \sigma_y$, so r indeed lies in $[-1, 1]$.) Substituting into (3):

$$\boxed{\hat{b} = r \frac{\sigma_y}{\sigma_x}} \quad (5)$$

This single line is the hub of the chapter; it says three things:

- (1) **The *strength* of correlation** ($|r|$, dimensionless, symmetric, unit-free) determines the *steepness* of the regression line in standard-deviation units: $\frac{\hat{b}\sigma_x}{\sigma_y} = r$. Moving x by one standard deviation moves the prediction $\hat{y}$ by r standard deviations.
- (2) **The symmetry of r vs. the directionality of regression:** r is unchanged under swapping x and y ; but the regression line $y \sim x$ (predicting y from x) and the regression line $x \sim y$ are two *different* lines (see Section 2.4).
- (3) **Goodness of fit:** the coefficient of determination

$$R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i (y_i - \bar{y})^2} = r^2, \quad (6)$$

i.e., the proportion of the sample variance of y explained linearly by x is exactly the square of the correlation coefficient. R^2 measures correlation, while r also carries a sign (direction).

Example 2.1 (Galton’s height regression). Galton (1886) studied the heights of 205 parent couples and 928 adult children, predicting child height y from mid-parent height x , and obtained a regression equation of approximately $y = 0.516x + 33.73$ (inches)[4]. The two generations have nearly identical standard deviations (about 3.3 inches), so in standard-deviation units the slope is simply $r \approx 0.5$: parents one standard deviation taller than average have children who are on average only half a standard deviation taller. Children of tall parents **regress toward mediocrity**, and so do children of short parents — hence the name “regression toward the mean.” This is not a law of biology; it is the arithmetic of (5): as long as $|r| < 1$ and each variable’s dispersion stays put, extremes necessarily fall back.

Remark. “Regression to the mean” is the ancestor of all statistical fallacies: the Sports Illustrated jinx, second-exam slump, and triumph-of-mediocrity arguments à la Secrist (1933) are all, at bottom, noise mistaken for signal. Galton himself avoided the trap with subtler reasoning[4, 11].

2.3 The variance viewpoint: where correlation comes from, and what it means

Look again through a different lens. Treat y in (1) as a random variable $y = bx + \varepsilon$, and assume $\text{Cov}(x, \varepsilon) = 0$ (i.e., x carries no component of y ’s noise — this *is itself a directional assumption!*). The algebra of variance gives

$$\text{Var}(y) = b^2 \text{Var}(x) + \text{Var}(\varepsilon). \quad (7)$$

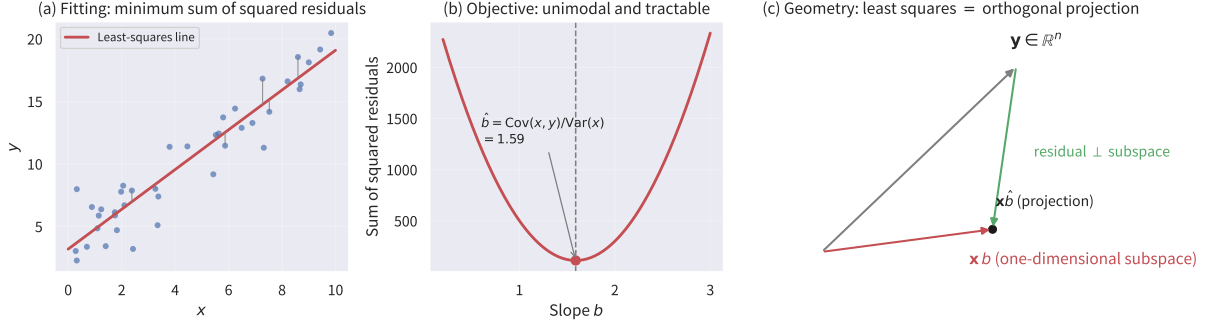


图 2: Three faces of one-dimensional least squares: (a) fitting — the squared sum of the residuals (vertical bars) is minimized; (b) optimization — the objective is unimodal in the slope b , and $\hat{b} = \text{Cov}(x, y)/\text{Var}(x)$ solves it in one step; (c) geometry — $\mathbf{x}\hat{b}$ is the orthogonal projection of $\mathbf{y}$ onto a one-dimensional subspace, with residuals $\perp$ to the subspace.

The fluctuation of y splits into two parts: the part carried by x , $b^2\text{Var}(x)$, and the irreducible noise $\text{Var}(\varepsilon)$. Immediately,

$$R^2 = \frac{b^2\text{Var}(x)}{\text{Var}(y)} = 1 - \frac{\text{Var}(\varepsilon)}{\text{Var}(y)}, \quad (8)$$

which converges with (6) from a different route: r^2 is simultaneously “goodness of fit at the sample level” and “the population share of explained variance.” This is the first payoff of the variance viewpoint — *correlation is, in essence, a transfer of variance*.

The second payoff is deeper. In (7), the sign of b describes “how fluctuations of x get translated into fluctuations of y .” But if the true structure is y causing x , and you regress $y \sim x$ anyway, you still obtain the same $\hat{b} \approx r\sigma_y/\sigma_x$. Correlation is *completely blind* to the direction of generation — the theme of the next section.

2.4 Directionality: two regression lines, and one major axis

Standardize both x and y (mean zero, variance one); then the **regression coefficients** in both directions equal r :

$$\text{coefficient of } y \sim x = r, \quad \text{coefficient of } x \sim y = r, \quad \text{product of the two} = r^2 < 1 \quad (|r| < 1).$$

(The second equality concerns the “coefficient,” not the “slope”: the regression of x on y solves to $x = ry$, which read with y as the independent variable indeed has coefficient r .) In other words, unless $|r| = 1$ (perfect linearity, no noise), **“the line that predicts y from x ” and “the line that predicts x from y ” are two different straight lines**. One terminological point must be made precise: *regression coefficients* and *geometric slopes drawn in the same coordinate system* are not the same thing. On the original scale, the two regression coefficients are respectively

$$b = r \frac{\sigma_y}{\sigma_x} \quad (y \sim x), \quad b' = r \frac{\sigma_x}{\sigma_y} \quad (x \sim y), \quad bb' = r^2 < 1,$$

— the product of the coefficients is r^2 , which is Galton’s “regression toward the mean” arithmetic; but if both lines are drawn in the same “ x horizontal, y vertical” coordinate system, the second must be rewritten as $y = \frac{\sigma_y}{r \sigma_x} x$, so the **geometric slopes** are $r \sigma_y / \sigma_x$ and $\sigma_y / (r \sigma_x)$, whose product is $(\sigma_y / \sigma_x)^2$ — identically 1 only when $\sigma_x = \sigma_y$ (including the standardized case). Both products are correct; they simply measure two different questions: the former asks “how much does each direction’s prediction shrink,” the latter asks “the opening angle between the two lines on the plot.” A bivariate-normal-ish point cloud approximates an ellipse, and the two regression lines are respectively: the line minimizing vertical residuals and the line minimizing horizontal residuals — neither coincides with the ellipse’s major axis (the major axis corresponds to PCA, which minimizes *perpendicular* distance). Burn Figure 3 into your mind, and “regression has direction, correlation does not” ceases to be a slogan.

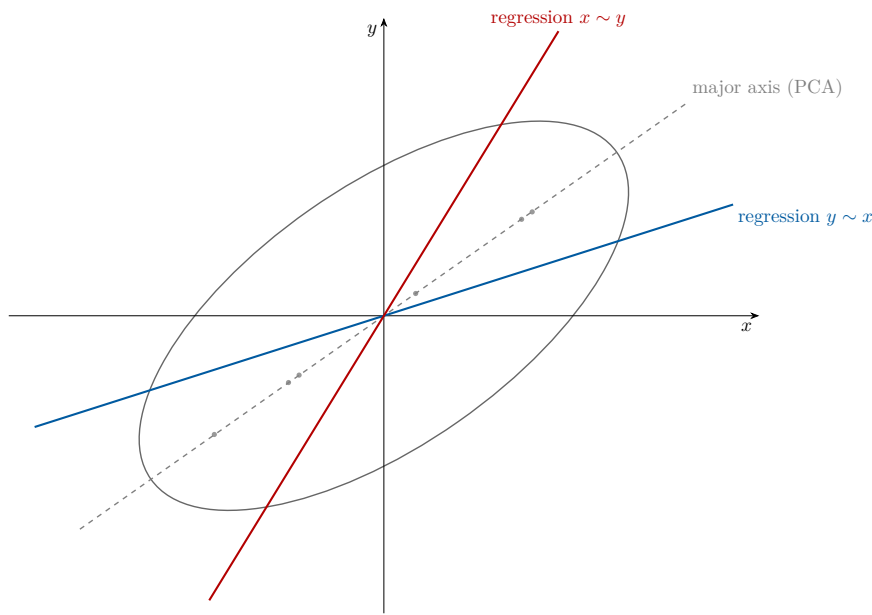


图 3: A standardized bivariate-normal point cloud (ellipse). Neither regression line coincides with the major axis: $y \sim x$ minimizes vertical residuals, $x \sim y$ minimizes horizontal residuals, and the major axis minimizes perpendicular residuals. Regression is a *directed* operation; the correlation coefficient r is not.

Directionality also has a purely algebraic statement: the decompositions $\text{Var}(y) = b^2 \text{Var}(x) + \text{Var}(\varepsilon)$ from (7) and $\text{Var}(x) = \tilde{b}^2 \text{Var}(y) + \text{Var}(\tilde{\varepsilon})$ (regressing x on y) can hold simultaneously, and each has $R^2 = r^2$. The two directions carry the same “explanatory power” — the variance expression of correlation’s symmetry; the slopes differ — the expression of directionality.

2.5 Causation: what a correlation coefficient can never give you

Why can’t r assert causation? Because at least three fundamentally different generative mechanisms produce *exactly the same* correlation coefficient:

Structure	Generative process	Observed correlation
x causes y	$y = bx + \varepsilon$	$r \neq 0$
y causes x	$x = \tilde{b}y + \tilde{\varepsilon}$	$r \neq 0$ (the same r)
Common cause z	$x = cz + u, y = dz + v$	$r = \text{sign}(cd) \cdot (\text{determined by } c, d)$

The data themselves have zero power to distinguish these three cases — a corollary of Reichenbach’s (1956) **common-cause principle**: if A and B are statistically correlated, then either A causes B , or B caused A , or there is a common cause C [7]. The principle provides the enumeration but not the decision — the decision must come from knowledge beyond the data.

Wright (1921) was the first to turn this into a calculus[6]. His **path analysis** says: on a given causal graph (whose arrows must be drawn beforehand from experiment, temporal order, or domain knowledge), the correlation between any two variables equals the sum, over all legal paths connecting them, of the products of the path coefficients. Hence:

- *Forward use*: given a causal hypothesis, use the observed correlations to *test* it (fit; residual correlations should vanish);
- *Reverse use*: inferring causal structure from correlations alone is beyond statistics — “the validity of these conclusions depends upon knowledge external to the method” (the substance of Wright’s original remark[6]).

More than a century later, this criterion remains the root of all causal inference (structural equation models, DAGs, do-calculus, difference-in-differences, instrumental variables): **regression outputs conditional association; reading a coefficient as a causal effect requires an identification strategy** — randomization, natural experiments, or the untestable “no unmeasured confounding” assumption. This boundary lies beyond the range of any t -statistic.

Takeaways of this chapter

- The one-dimensional least-squares slope $\hat{b} = \text{Cov}_n(x, y) / \text{Var}_n(x) = r \sigma_y / \sigma_x$: regression generalizes the mean, the correlation coefficient is its standardized form, and $R^2 = r^2$ is both goodness of fit and the share of explained variance.
- Variance viewpoint: $\text{Var}(y) = b^2 \text{Var}(x) + \text{Var}(\varepsilon)$ — correlation is the transfer of variance; $\text{Cov}(x, \varepsilon) = 0$ is already a directional assumption.
- Directionality: $y \sim x$ and $x \sim y$ are two different regression lines (the product of regression coefficients is r^2 ; the product of geometric slopes drawn in one coordinate system is $(\sigma_y / \sigma_x)^2$, equal to 1 only when $\sigma_x = \sigma_y$); correlation is symmetric, regression is not.
- Causation: at least three structures produce the same r ; the common-cause principle enumerates them, path analysis provides “a calculus given external causal knowledge,” but the data alone can never decide the direction.

3 High-Dimensional Linear Regression: The Closed-Form World When $n \gg d$

One dimension builds intuition; high dimensions do the work. This chapter considers a sample of size n and feature dimension d , with the model

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \text{i.e.} \quad \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (9)$$

where $\mathbf{X} = [\mathbf{x}_1^\top; \dots; \mathbf{x}_n^\top] \in \mathbb{R}^{n \times d}$ is the design matrix, $\mathbf{y} \in \mathbb{R}^n$, and $\boldsymbol{\beta} \in \mathbb{R}^d$ is unknown. We first derive the closed-form solution in the classical **overdetermined** regime $n \gg d$, then read off its geometric and statistical meaning, and finally watch what breaks first as n approaches d — and how to fix it.

3.1 Least Squares and the Normal Equations

The total residual sum of squares is

$$S(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}. \quad (10)$$

Using the vector calculus identities $\nabla_{\boldsymbol{\beta}}(\mathbf{c}^\top \boldsymbol{\beta}) = \mathbf{c}$ and $\nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top \mathbf{A} \boldsymbol{\beta}) = 2\mathbf{A}\boldsymbol{\beta}$ (for symmetric $\mathbf{A}$), the gradient is

$$\nabla_{\boldsymbol{\beta}} S = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}. \quad (11)$$

Setting it to zero yields the **normal equations**:

$$\boxed{\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}} \quad (12)$$

3.2 The Closed-Form Solution for $n \gg d$: Existence, Uniqueness, Computability

Theorem 3.1 (Closed-form solution and uniqueness). *If $\mathbf{X}$ has full column rank, i.e. $\text{rank}(\mathbf{X}) = d$ (equivalently, its d columns are linearly independent, which requires $n \geq d$), then the Gram matrix $\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{d \times d}$ is symmetric positive definite and invertible, and the normal equations have the unique solution*

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}} \quad (13)$$

Moreover, since the Hessian $\nabla^2 S = 2\mathbf{X}^\top \mathbf{X} \succ \mathbf{0}$, S is strictly convex and (13) is the unique global minimizer of S .

证明. For any $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 > 0$: if it were 0 then $\mathbf{X}\mathbf{v} = \mathbf{0}$, contradicting linear independence of the columns. Hence $\mathbf{X}^\top \mathbf{X} \succ \mathbf{0}$; a positive definite matrix is invertible (all its eigenvalues are positive), and substitution gives (13). A stationary point of a strictly convex function is its unique global minimizer. $\square$

Note what the condition really says. “ $n \gg d$ ” is merely colloquial for “ample sampling”; the theorem requires only **full column rank** —it guarantees that the map from β to the fitted values $\mathbf{X}\beta$ has no redundant direction, so the parameters are uniquely determined by the data. In practice $n \gg d$ makes full column rank *highly likely* (a random design with a continuous distribution is almost surely of full rank); this is the default working regime of classical statistics.

For completeness, the rank-deficient case: if some column is an exact linear combination of the others (perfect collinearity), $\mathbf{X}^\top \mathbf{X}$ is singular and $\hat{\beta}$ is not unique —but the fitted values $\mathbf{X}\hat{\beta}$ still are. All minimum-norm solutions are given uniformly by the **Moore–Penrose pseudoinverse** $\mathbf{X}^+$ [8]:

$$\hat{\beta} = \mathbf{X}^+ \mathbf{y}, \quad (14)$$

which is exactly what R’s `lm`, NumPy’s `lstsq`, and scikit-learn’s `LinearRegression` compute (via QR/SVD factorizations rather than explicit inversion; see Section 3.7).

3.3 Geometry: A Projection, Again

Think of the columns of $\mathbf{X}$ as d vectors in $\mathbb{R}^n$; their column space

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^d\} \subseteq \mathbb{R}^n$$

is the set of **all fitted-value vectors the model can produce**. Fitting the model means finding the point of $\mathcal{C}(\mathbf{X})$ closest to $\mathbf{y}$, and Euclidean distance is exactly $\|\mathbf{y} - \mathbf{X}\beta\|$ —the square root of the loss (10). Hence:

The geometric statement of least squares

$$\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\| \iff \mathbf{X}\hat{\beta} \text{ is the orthogonal projection of } \mathbf{y} \text{ onto } \mathcal{C}(\mathbf{X}).$$

The normal equations $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{0}$ say: **the residual vector is orthogonal to every column of $\mathbf{X}$** (“normal” is geometry-speak for “perpendicular”). By Pythagoras, for any other $\mathbf{X}\beta \in \mathcal{C}(\mathbf{X})$,

$$\|\mathbf{y} - \mathbf{X}\beta\|^2 = \|\mathbf{X}(\hat{\beta} - \beta)\|^2 + \|e\|^2 \geq \|e\|^2,$$

with equality if and only if $\beta = \hat{\beta}$ —a proof with no calculus: perpendicularity implies optimality.

This yields a matrix that runs through all of regression computation. The **hat matrix**

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top, \quad \hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad e = (\mathbf{I} - \mathbf{H})\mathbf{y}. \quad (15)$$

Verify directly: $\mathbf{H}^\top = \mathbf{H}$ and $\mathbf{H}^2 = \mathbf{H}$ —**symmetric + idempotent = orthogonal projection matrix**. Two consequences ready for use: $\text{tr}(\mathbf{H}) = \text{tr}((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}) = d$ (the origin of the d degrees of freedom, the residuals consuming $n - d$); the diagonal entries $H_{ii} \in [0, 1]$ measure the **leverage** of observation i (an outlying $\mathbf{x}$ pushes H_{ii} up —the starting point of regression diagnostics). All one-dimensional conclusions generalize: fitted values and residuals are orthogonal, and R^2 is still the “fraction of variance explained.”

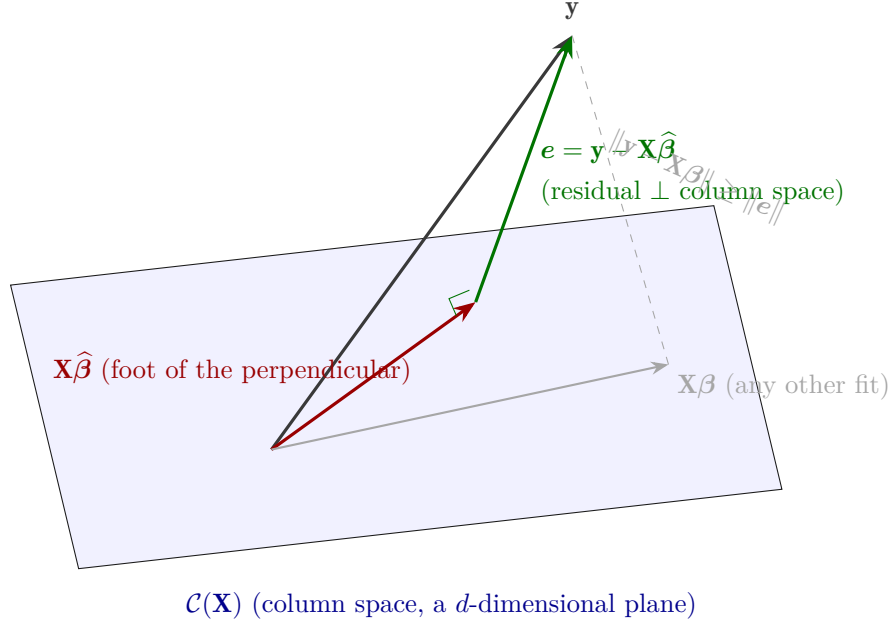


图 4: The geometry of least squares: the foot of the perpendicular from $\mathbf{y}$ to the column space $\mathcal{C}(\mathbf{X})$ is $\mathbf{X}\hat{\boldsymbol{\beta}}$. The vertical segment $\mathbf{e}$ is the shortest residual (green); any other fit $\mathbf{X}\boldsymbol{\beta}$ (gray) reaches $\mathbf{y}$ along a longer slanted line —perpendicularity is optimality.

3.4 Statistics: When Is This Closed-Form Solution “Optimal”?

The closed form only promises “closest to the data.” Statistics asks: as an *estimator* of the unknown truth $\boldsymbol{\beta}$, how good is it? The **Gauss–Markov theorem**, proved by Gauss (1823) and later formalized by Markov (1912), answers[3]:

Gauss–Markov, informal statement

Under model (9), if the errors satisfy: zero mean ($\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$, i.e. correct model specification), homoskedastic and uncorrelated ($\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$), and $\mathbf{X}$ has full column rank —**without assuming any error distribution**—then the least squares estimator $\hat{\boldsymbol{\beta}}$ has the smallest variance among all estimators that are linear in $\mathbf{y}$ and unbiased: it is the *Best Linear Unbiased Estimator* (BLUE).

Proof skeleton (the full argument is Exercise 3.1): write any linear unbiased competitor as $\tilde{\boldsymbol{\beta}} = (\mathbf{B} + \mathbf{D})\mathbf{y}$ with $\mathbf{B} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$; unbiasedness forces $\mathbf{D}\mathbf{X} = \mathbf{0}$. Then

$$\text{Var}(\tilde{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{B} + \mathbf{D})(\mathbf{B} + \mathbf{D})^\top = \underbrace{\sigma^2 \mathbf{B}\mathbf{B}^\top}_{\text{Var}(\hat{\boldsymbol{\beta}})} + \underbrace{\sigma^2 \mathbf{D}\mathbf{D}^\top}_{\succeq \mathbf{0}},$$

the cross terms vanishing because $\mathbf{D}\mathbf{X} = \mathbf{0}$. In one sentence: *any unbiased deviation from OLS is a zero-mean perturbation uncorrelated with OLS, and an uncorrelated perturbation can only add variance.*

Two qualifiers mark the theorem’s boundaries:

- **“Linear”**: if one further assumes Gaussian errors, the Cramér–Rao lower bound shows that *no* unbiased estimator whatsoever (linear or not) can beat $\hat{\beta}$;
- **“Unbiased”**: once bias is permitted (ridge, lasso), smaller mean squared error becomes available —see Section 3.6.

3.5 Why Squared Error, of All Losses?

Question (3) from the introduction can now be answered head-on. What exactly is lost by switching to another loss? Three progressively deeper answers.

- (1) **The geometric–optimization answer**: squaring makes the loss quadratic —differentiable, strictly convex, stationary point equals global optimum, and the normal equations are a *linear* system; Chapter 4 will further show that it gives gradient descent an exact closed-form solution in t . Switch to absolute-value loss: still convex, but nondifferentiable at zero, the optimum is a median rather than a mean, and the “normal equations” become a set of piecewise-linear conditions —the same door opens into a far more complicated room.
- (2) **The probabilistic answer** (Gauss, 1809): if $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ i.i.d., the likelihood of the data is proportional to

$$\exp\left(-\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\beta\|^2\right),$$

so maximizing the likelihood $\iff$ minimizing the sum of squared residuals —**squared error is exactly maximum likelihood under Gaussian noise**[2]. It was along this route that Gauss gave least squares its probabilistic foundation, arguing that “nature measures errors by the bell curve.” This answer confers an **hypothesis identity** on the loss: squared error asserts that “large errors become exponentially-squared unlikely,” a far more aggressive claim than the Laplace assumption behind median regression (exponential-linear decay) —and outliers pull the squared-error solution correspondingly harder.

- (3) **The statistical answer**: it *is* Gauss–Markov —under the *weak* conditions of zero mean, homoskedasticity, and uncorrelatedness, $\hat{\beta}$ is already optimal among all linear unbiased estimators in the minimum-variance sense; with Gaussian errors, Cramér–Rao upgrades this to optimality among *all* unbiased estimators.

One-sentence summary: *squared error is the shadow cast by the assumption pair “errors approximately Gaussian + estimator required to be linear and unbiased.”* Changing the loss (absolute, Huber, quantile) is never “just a different metric” —it is a different theory of the error world. Audit the assumptions first, then talk about changing the loss; that is precisely the entry point of robust regression and generalized linear models.

Exercise 3.1. Complete the proof of the Gauss–Markov theorem: show that any linear unbiased estimator $\tilde{\beta} = \mathbf{A}\mathbf{y}$ can be written as $(\mathbf{B} + \mathbf{D})\mathbf{y}$ (with $\mathbf{B} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$) where $\mathbf{D}\mathbf{X} = \mathbf{0}$, and derive $\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}) = \sigma^2 \mathbf{D}\mathbf{D}^\top \succeq \mathbf{0}$.

3.6 When n No Longer Dominates d : What Breaks, and How to Fix It

The Gauss–Markov world is entirely an $n \gg d$ world. Now push n toward d , in order of breakage:

- (1) $d \rightarrow n$: **$\mathbf{X}^\top \mathbf{X}$ becomes ill-conditioned.** Some feature directions are barely supported by the data (small eigenvalues), and $(\mathbf{X}^\top \mathbf{X})^{-1}$ amplifies noise precisely along them —the variance of $\hat{\boldsymbol{\beta}}$ explodes (the diagonal entries of the covariance $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ blow up), the *signs* of coefficients can be unstable, while predictions at the observed points remain fine. The disease lives in parameter space, not in prediction space.
- (2) $d = n$: **$\mathbf{X}$ is square and invertible, residuals are zero.** Fitting becomes interpolation: training error 0, generalization error takes off —the algebraic face of overfitting.
- (3) $d > n$: **$\mathbf{X}^\top \mathbf{X}$ is necessarily singular.** Infinitely many solutions exist; OLS is *undefined*, and external force must pick one.

The fixes share one idea: impose a “size budget” on the parameters. **Ridge regression** (Hoerl & Kennard, 1970)[9]:

$$\hat{\boldsymbol{\beta}}^{\text{ridge}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2 \implies \hat{\boldsymbol{\beta}}^{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (16)$$

Three perspectives: **numerically**, $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I} \succ \mathbf{0}$ always —any $\lambda > 0$ restores uniqueness; **geometrically**, it is the Lagrange form of the constrained problem $\min \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ s.t. $\|\boldsymbol{\beta}\|^2 \leq c$, shrinking coefficients along the ill-conditioned directions; **statistically**, via the SVD $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, ridge replaces each $1/d_j$ of OLS by $d_j/(d_j^2 + \lambda) < 1/d_j$ —*taming exactly the small-singular-value directions that OLS amplifies*. The cost is bias (Exercise 3.2); the gain is a large reduction in variance, so the total mean squared error actually decreases in the weak-signal regime. **Lasso** (Tibshirani, 1996) swaps in the ℓ_1 penalty $\lambda \|\boldsymbol{\beta}\|_1$, driving some coefficients to exactly 0 and performing variable selection on the way[10]. The penalty λ is chosen by cross-validation, which estimates prediction error —and the loop closes: *regularization is not the opposite of OLS; it is OLS’s survival plan when $d \approx n$.*

Exercise 3.2. Derive the expression for $\hat{\boldsymbol{\beta}}^{\text{ridge}}$ from (16) and show that its bias is $-\lambda(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \boldsymbol{\beta}$.

3.7 A Computational Note: What Proofs Write vs. What Engineers Run

In *proofs* we write $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$; in *computation* nobody ever inverts —forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number of $\mathbf{X}$ and is numerical self-sabotage. The engineering route is the QR factorization $\mathbf{X} = \mathbf{Q}\mathbf{R}$ ($\mathbf{Q}$ with orthonormal columns, $\mathbf{R}$ upper triangular), giving

$$\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top, \quad \hat{\boldsymbol{\beta}} = \mathbf{R}^{-1} \mathbf{Q}^\top \mathbf{y} \text{ (back-substitution; never an explicit inverse).}$$

$\mathbf{Q}^\top \mathbf{y}$ is exactly the projection coordinate of $\mathbf{y}$ in an orthogonal basis —the geometry of Section 3.3 is what runs on the computer. **The proof’s formula and the computation’s algorithm**

are two faces of one object: another standard answer to “why study linear regression” —it is the only model where both faces can be seen clearly at the same time.

Chapter takeaways

- High-dimensional least squares has the unique closed-form solution $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ whenever $\mathbf{X}$ has full column rank (almost surely when $n \gg d$); strict convexity makes it the global optimum;
- Geometrically it is the orthogonal projection of $\mathbf{y}$ onto the column space; the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is symmetric and idempotent, with $\text{tr}(\mathbf{H}) = d$ degrees of freedom;
- Gauss–Markov: under zero-mean, homoskedastic, uncorrelated errors, OLS is BLUE —optimality with zero distributional assumptions; with Gaussian errors, upgraded to optimal among *all* unbiased estimators;
- As $d \rightarrow n$ the variance explodes first (ill-conditioning), and at $d \geq n$ the solution ceases to be unique (singularity); ridge $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ and lasso are the regularized fixes of the same idea, with λ chosen by cross-validation;
- In practice QR/SVD replaces explicit inversion; proof and computation are two faces of one geometry.

4 Gradient Descent, the Optimization View: Iteration, Fixed Points, and an Exact Solution

Section 3.2 gave the closed-form solution $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, but “can be written down” is not “should be computed that way”: explicit inversion solves a $d \times d$ linear system at cost $O(d^3)$, and forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number of $\mathbf{X}$. This chapter switches to the optimization view: instead of solving the normal equations directly, start from some $\hat{\beta}_0$ and let the loss function “walk downhill.” We will see —**because the loss is quadratic, this downhill path can be written down exactly**: $\hat{\beta}_t$ has a closed-form expression in t , $\mathbf{X}$, and $\mathbf{y}$, and its fixed point is precisely the closed-form solution $\hat{\beta}$.

4.1 The Gradient Descent Iteration and Its Fixed Point

Adopt the standard convention from machine learning:

$$L(\beta) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2, \quad \nabla L(\beta) = \mathbf{X}^\top (\mathbf{X}\beta - \mathbf{y}), \quad (17)$$

(the $\frac{1}{2}$ is only a convention so that the gradient carries no factor 2; if you insist on $L = \|\cdot\|^2$, replace every η below by 2η .) Note that the iteration as first written,

$$\hat{\beta}_{t+1} = \hat{\beta}_t - \eta L(\hat{\beta}_t),$$

needs a small correction: L is a scalar, and a scalar cannot be subtracted from a vector —the downhill direction is the **negative gradient**, with step size (learning rate) $\eta > 0$:

$$\boxed{\hat{\beta}_{t+1} = \hat{\beta}_t - \eta \nabla L(\hat{\beta}_t) = \hat{\beta}_t - \eta \mathbf{X}^\top (\mathbf{X}\hat{\beta}_t - \mathbf{y})} \quad (18)$$

A **fixed point** is one that stops moving after an iteration step:

$$\beta^* = \beta^* - \eta \nabla L(\beta^*) \iff \nabla L(\beta^*) = \mathbf{0} \iff \mathbf{X}^\top \mathbf{X} \beta^* = \mathbf{X}^\top \mathbf{y}. \quad (19)$$

The fixed-point condition *is* the normal equation! So when $\mathbf{X}$ has full column rank and $0 < \eta < \infty$, (18) has the unique fixed point

$$\boxed{\beta^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \hat{\beta}} \quad (20)$$

Gradient descent and least squares are two paths to the same object: *one solves the equations directly (Section 3.2), the other iterates toward them (this section) —and they share the same destination.* This is the first layer of “putting the two formulas together.”

4.2 Putting the Two Formulas Together: An Exact Solution for $\hat{\beta}_t$

Now for the second layer: combine (18) with (20) to write the **exact solution at any time t** .

Rewrite (18) in “error propagation” form. Let $\mathbf{A} := \mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}$; the fixed point gives

$$\eta \mathbf{X}^\top \mathbf{y} = \eta \mathbf{X}^\top \mathbf{X} \beta^* = (\mathbf{I} - \mathbf{A}) \beta^*.$$

Substituting into (18):

$$\hat{\beta}_{t+1} = (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}) \hat{\beta}_t + \eta \mathbf{X}^\top \mathbf{y} = \mathbf{A} \hat{\beta}_t + (\mathbf{I} - \mathbf{A}) \beta^*,$$

i.e.

$$\hat{\beta}_{t+1} - \beta^* = \mathbf{A} (\hat{\beta}_t - \beta^*).$$

The error vector is multiplied by the same matrix $\mathbf{A}$ at every step —the entire secret of linear iteration lives in this line. Unrolling t times:

Theorem 4.1 (Exact solution for $\hat{\beta}_t$). *For any initialization $\hat{\beta}_0$ and any $t \geq 0$,*

$$\hat{\beta}_t = \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}}_{\hat{\beta}} + (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t (\hat{\beta}_0 - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}) \quad (21)$$

“closed-form solution + t -th power decay of the initial error.” In particular, with the most common default $\hat{\beta}_0 = \mathbf{0}$:

$$\hat{\beta}_t = \left[\mathbf{I} - (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t \right] (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (22)$$

证明. (21) is the expansion of $\hat{\beta}_t - \beta^* = \mathbf{A}^t (\hat{\beta}_0 - \beta^*)$, which follows by inducting the error recursion above over t steps (for $t = 0$, $\mathbf{A}^0 = \mathbf{I}$ is trivial). For (22), substitute $\hat{\beta}_0 = \mathbf{0}$ and use $\beta^* = \hat{\beta}$. $\square$

This exact solution makes three things transparent:

- (1) **Whether convergence happens at all is decided by the eigenvalues of $\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}$.**

Let $\lambda_1 \geq \dots \geq \lambda_d > 0$ be the eigenvalues of $\mathbf{X}^\top \mathbf{X}$ (full column rank makes them all positive); then the eigenvalues of $\mathbf{A}$ are $1 - \eta \lambda_j$, and the error decays along each eigen-direction $\mathbf{v}_j$ by the factor $(1 - \eta \lambda_j)^t$.

- (2) **Convergence to $\hat{\beta}$ as $t \rightarrow \infty$ happens if and only if every direction contracts:**

$$|1 - \eta \lambda_j| < 1 \quad \forall j \quad \iff \quad 0 < \eta < \frac{2}{\lambda_{\max}(\mathbf{X}^\top \mathbf{X})} \quad (23)$$

If the step is too large ($\eta \geq 2/\lambda_{\max}$), the largest-eigenvalue direction diverges and the iteration explodes —“the learning rate must not be too large” acquires its precise algebraic meaning here for the first time.

- (3) **Gradient descent is “inverting approximately.”** When $\eta < 1/\lambda_{\max}$, $\|\mathbf{A}\| < 1$, and the geometric (Neumann) series applies:

$$\eta \sum_{k=0}^{t-1} \mathbf{A}^k = \eta \sum_{k=0}^{t-1} (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^k \xrightarrow{t \rightarrow \infty} (\mathbf{X}^\top \mathbf{X})^{-1},$$

so (22) is precisely “approximating $(\mathbf{X}^\top \mathbf{X})^{-1}$ by a degree- t polynomial.” The closed form obtains the inverse in one shot; gradient descent assembles the inverse piece by piece through iteration —the same $(\mathbf{X}^\top \mathbf{X})^{-1}$, one delivered outright, the other on an installment plan.

4.3 The Optimal Step Size, and the Shadow of the Condition Number

Within the convergence interval (23), the slowest direction sets the pace: the asymptotic convergence rate is $\max_j |1 - \eta\lambda_j|$. The optimal step equalizes the decay factors at the two ends:

Corollary 4.2 (Optimal step size and convergence rate). *Let $\lambda_{\min}, \lambda_{\max}$ be the smallest and largest eigenvalues of $\mathbf{X}^\top \mathbf{X}$, and $\kappa = \lambda_{\max}/\lambda_{\min}$ the condition number. The optimal step size and the corresponding asymptotic convergence rate are*

$$\eta^* = \frac{2}{\lambda_{\min} + \lambda_{\max}}, \quad \rho^* = \max_j |1 - \eta^* \lambda_j| = \frac{\kappa - 1}{\kappa + 1} < 1. \quad (24)$$

证明. $\max_j |1 - \eta\lambda_j| = \max\{|1 - \eta\lambda_{\min}|, |1 - \eta\lambda_{\max}|\}$. Within the convergence interval the former decreases in η and the latter increases, so the optimum occurs where $1 - \eta\lambda_{\min} = \eta\lambda_{\max} - 1$; solving gives η^* , and substitution gives $\rho^* = (\lambda_{\max} - \lambda_{\min})/(\lambda_{\max} + \lambda_{\min})$. $\square$

The three regimes of the step size. Classify η by its relation to the spectral endpoints $1/\lambda_{\max}$ and $2/\lambda_{\max}$; the three regimes look entirely different on the *trajectory* (Figure 6) and in the signed error (Figure 5):

Regime 1: $0 < \eta \leq 1/\lambda_{\max}$: **monotone descent.** Now $0 \leq 1 - \eta\lambda_j < 1$ for every eigenvalue: every direction's error factor is nonnegative, the iterate approaches $\hat{\beta}$ along a straight line, the signed error never changes sign, and the loss never looks back (Figure 5(a)). This regime is the safest but the slowest —measured along the smallest-eigenvalue direction, each step compresses by only $1 - \eta\lambda_{\min}$; the large step is locked out by the largest eigenvalue.

Regime 2: $1/\lambda_{\max} < \eta < 2/\lambda_{\max}$: **oscillating trajectory, still descending loss.** Once the step crosses into the second regime, the factor $1 - \eta\lambda_{\max}$ on the largest-eigenvalue direction turns negative: the iterate *overshoots and flips sign* across the valley floor every step —the zigzag track in Figure 6(b) and the sawtooth of the signed error in Figure 5(b). But note a fact that is often missed: *the loss itself does not oscillate*. From (21),

$$L(\hat{\beta}_t) - L_{\min} = \frac{1}{2} \sum_{j=1}^d \lambda_j (1 - \eta\lambda_j)^{2t} c_j^2, \quad c_j = \mathbf{v}_j^\top (\hat{\beta}_0 - \hat{\beta}),$$

where every term carries an *even* power —sign flips are killed by the squaring, and the loss curve keeps sliding down monotonically (the gray dashed line in Figure 5(b)).

What oscillates is the trajectory, not the loss. As long as $|1 - \eta\lambda_{\max}| < 1$ the sawtooth amplitude keeps decaying —oscillation is not divergence. Curiously, the optimal step $\eta^* = 2/(\lambda_{\min} + \lambda_{\max})$ falls in this regime (when $\kappa > 1$, $1/\lambda_{\max} < \eta^* < 2/\lambda_{\max}$): *the fastest strategy is not to never look back, but to allow a controlled overshoot along the longest direction, trading flips for the fastest overall decay.*

Regime 3: $\eta \geq 2/\lambda_{\max}$: **oscillation without convergence**. The factor $|1 - \eta\lambda_{\max}| \geq 1$: the iterate flips sign on the largest-eigenvalue direction every step without any decay in amplitude (equality leaves the sawtooth alive forever) —a diverging zigzag (Figure 6(c)), with the signed error and the loss exploding together within a handful of steps (Figure 5(c)).

Both boundaries, $1/\lambda_{\max}$ and $2/\lambda_{\max}$, depend only on the largest eigenvalue —the precise algebraic explanation of the engineering experience “set the learning rate too large and you get NaN.” And within the regimes, the trade-off between speed and stability is adjudicated by the condition number κ .

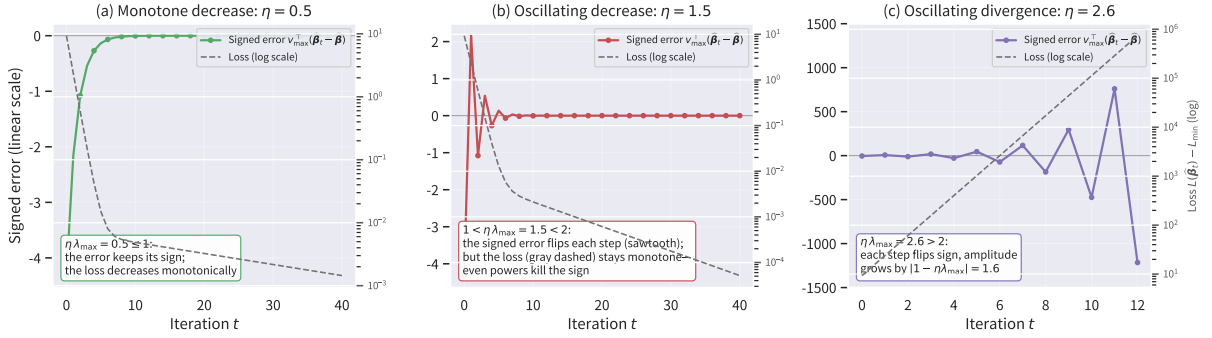


图 5: The three regimes of the signed error (colored solid lines, left axis, linear scale) and the loss (gray dashed, right axis, log scale) versus iteration count t —same quadratic problem, same initialization, only η varies: (a) monotone descent — $\eta\lambda_{\max} \leq 1$, the largest-eigenvalue signed error never changes sign and the loss is monotone; (b) oscillating descent — $1 < \eta\lambda_{\max} < 2$, the signed error flips sign every step into a sawtooth, yet the loss, being an even power of the error factors, keeps decreasing monotonically; (c) oscillation without convergence — $\eta\lambda_{\max} > 2$, the error flips sign every step with amplitude growing by $|1 - \eta\lambda_{\max}|$, and the loss explodes.

Two practical corollaries, both fragments of the answer to “why study linear regression”:

- **The condition number rules everything.** When $\kappa \approx 1$ (eigen-directions of similar length), $\rho^* \approx 0$ and convergence takes a handful of steps; when $\kappa \gg 1$ (the ill-conditioning of Section 3.6), $\rho^* \approx 1 - 2/\kappa$ and convergence needs $O(\kappa \log(1/\varepsilon))$ steps —the engineering folklore “standardize your features first” is here half a theorem: standardization normalizes the diagonal of $\mathbf{X}^\top \mathbf{X}$ and **removes the part of the condition number due to units** — but eigenvalue spread caused by **correlation structure** (collinearity) is untouched, and κ can still be large.
- **Preconditioning = GD in better coordinates.** Whitening after a spectral decomposition of $\mathbf{X}^\top \mathbf{X}$ is equivalent to running gradient descent in a better coordinate system; conjugate gradients (CG) can be viewed as “GD with an automatically chosen step size and direction at every step,” reaching $\hat{\beta}$ **exactly** in d steps (finite termination for quadratic functions). From GD to CG to direct inversion is one family: the two ends are “iterate

t steps” and “one closed-form step,” and everything in between is an interpolation of the same geometry.

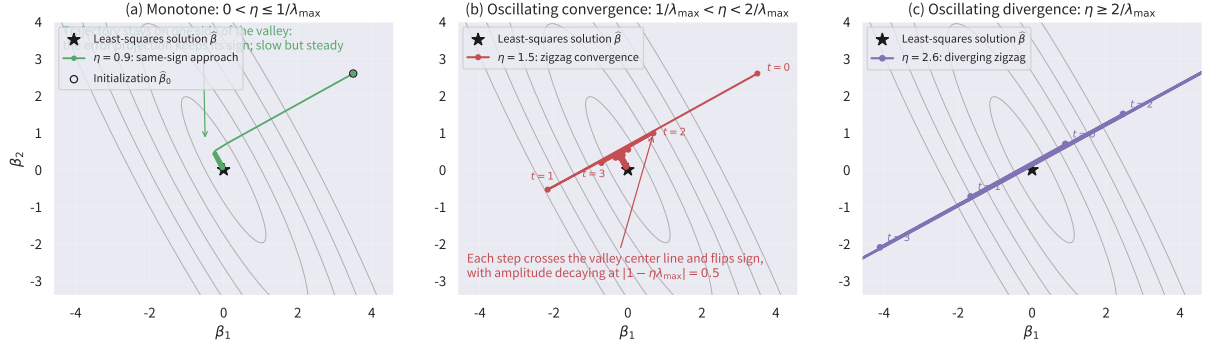


图 6: Three trajectories of gradient descent on a quadratic loss (an elliptical valley with condition number $\kappa = 25$), one per step-size regime: (a) $\eta = 0.9 \leq 1/\lambda_{\max}$ —the trajectory never crosses the valley floor, approaching without sign flips, but crawls slowly along the “thin” direction; (b) $1/\lambda_{\max} < \eta = 1.5 < 2/\lambda_{\max}$ —the trajectory crosses the valley floor and flips sign every step, the zigzag amplitude decaying by $|1 - \eta\lambda_{\max}| = 0.5$ —oscillating descent; (c) $\eta = 2.6 > 2/\lambda_{\max}$ —every step flips sign with amplitude growing by 1.6, flying out of view within a few steps.

4.4 A Side-by-Side Comparison with the Closed-Form Solution

	Closed form $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$	Gradient descent (18)
Character	One shot (solves a $d \times d$ system)	Successive approximation, error $\propto \rho^t$ after t steps
Cost (full column rank)	$O(nd^2 + d^3)$; must form $\mathbf{X}^\top \mathbf{X}$	$O(nd)$ matrix-vector products per step; memory-friendly
Applicable regime	The classical region $n \gg d$	Any n, d (with regularization / early stopping)
Ill-conditioning $\kappa \gg 1$	Numerically unstable (squared condition number)	Painfully slow convergence (the same κ , a different disease)
Relation to $\hat{\beta}$	It <i>is</i> $\hat{\beta}$	Its fixed point is $\hat{\beta}$; $\hat{\beta}_t$ has the exact solution (21)

Chapter takeaways

- The fixed-point condition $\nabla L = \mathbf{0}$ of the iteration $\hat{\beta}_{t+1} = \hat{\beta}_t - \eta \mathbf{X}^\top (\mathbf{X} \hat{\beta}_t - \mathbf{y})$ is the normal equation; the unique fixed point is the closed-form solution $\hat{\beta}$ —the optimization and the algebraic views end at the same place;
- Combining the two formulas yields the exact solution $\hat{\beta}_t = \hat{\beta} + (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t (\hat{\beta}_0 - \hat{\beta})$: the error is multiplied by $\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}$ each step, and the eigenvalues $1 - \eta \lambda_j$ govern the decay direction by direction;
- Convergence requires $0 < \eta < 2/\lambda_{\max}$; the optimal $\eta^* = 2/(\lambda_{\min} + \lambda_{\max})$ has rate $(\kappa - 1)/(\kappa + 1)$ —“standardize your features” removes unit-driven ill-conditioning (correlation-driven ill-conditioning it cannot fix);
- The geometric-series form shows GD assembling $(\mathbf{X}^\top \mathbf{X})^{-1}$ by a degree- t polynomial: the closed form hands over the inverse, the iteration pays it in installments; conjugate

gradients is the midpoint of the family.

Exercise 4.1. From (21), derive: if $\hat{\beta}_0 = \mathbf{0}$ and $\eta = \eta^*$, then $\|\hat{\beta}_t - \hat{\beta}\| \leq \left(\frac{\kappa-1}{\kappa+1}\right)^t \|\hat{\beta}\|$; and from this, give a big- O expression for the number of steps needed to reach $\|\hat{\beta}_t - \hat{\beta}\| \leq \varepsilon$.

Exercise 4.2. Verify the identity $\eta \sum_{k=0}^{t-1} (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^k \mathbf{X}^\top \mathbf{y} = [\mathbf{I} - (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t] \hat{\beta}$ (hint: multiply both sides by $\mathbf{X}^\top \mathbf{X}$, or use the Neumann series), and explain why it is precisely the left-multiplied form of (22).

5 The Other Fixed Point: Where Gradient Descent Goes When $d > n$

The exact solution (21) of Chapter 4 has a hidden premise: $\mathbf{X}^\top \mathbf{X}$ invertible. This chapter turns to the opposite, **underdetermined** regime $d > n$: fewer equations than unknowns, so $\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{d \times d}$ is necessarily singular ($\text{rank} \leq n < d$), the closed form (13) no longer exists, and the decay matrix $\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}$ of Chapter 4 has $d - n$ eigenvalues identically equal to 1 —the error **never decays** along certain directions. Yet it is precisely here that gradient descent delivers a profound result: **the iteration still converges; it converges to a special solution selected by the initialization; and starting from $\hat{\beta}_0 = \mathbf{0}$, it converges to the minimum-norm solution**

$$\hat{\beta}' = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y}.$$

5.1 Setup: $d > n$, and the Solution Set Is a Manifold

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ have **full row rank** ($\text{rank}(\mathbf{X}) = n$: the n rows are linearly independent, as continuously distributed random data almost surely are). Then $\mathbf{X} \mathbf{X}^\top \in \mathbb{R}^{n \times n}$ is symmetric positive definite and invertible. In this case:

- The system $\mathbf{X}\beta = \mathbf{y}$ is **consistent** ($\mathbf{y}$ necessarily lies in the column space), and its solutions are **not unique**: the general solution is

$$\{\beta : \mathbf{X}\beta = \mathbf{y}\} = \hat{\beta}' + \mathcal{N}(\mathbf{X}), \quad \mathcal{N}(\mathbf{X}) = \{\mathbf{w} \in \mathbb{R}^d : \mathbf{X}\mathbf{w} = \mathbf{0}\},$$

a $(d - n)$ -dimensional affine subspace (the solution manifold) —every point of it is an interpolating solution;

- The loss $L(\beta) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2$ vanishes identically on this manifold: the global minimum is 0 and is attained **on a manifold**;
- $\mathbf{X}^\top \mathbf{X}$ has $d - n$ zero eigenvalues, and $(\mathbf{X}^\top \mathbf{X})^{-1}$ is undefined —all the formulas of Chapter 4 fail here.

5.2 A New Fixed Point, and Two Verifications

Verification 1: exact interpolation

Substitute directly:

$$\mathbf{X} \hat{\beta}' = \mathbf{X} \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y} = \mathbf{y}. \quad (25)$$

That is, $\mathbf{y} = \mathbf{X} \hat{\beta}'$: training error exactly zero —**interpolation**. This is the continuation, into $d > n$, of the “zero residuals” phenomenon at $d = n$ (Section 3.6): excess capacity turns fitting into memorization.

Verification 2: a first-order equilibrium

The gradient is

$$\nabla L(\hat{\beta}') = \mathbf{X}^\top (\mathbf{X}\hat{\beta}' - \mathbf{y}) = \mathbf{X}^\top (\mathbf{y} - \mathbf{y}) = \mathbf{0}. \quad (26)$$

So $\hat{\beta}'$ is a **first-order equilibrium** (stationary point) of the least-squares loss; since L is convex and $L(\hat{\beta}') = 0 = \min L$, it is in fact a global minimum. Hence it is also a fixed point of the GD iteration (18): $\hat{\beta}' - \eta \nabla L(\hat{\beta}') = \hat{\beta}'$ for **every** η . Indeed (26) shows more: **every point of the entire solution manifold $\hat{\beta}' + \mathcal{N}(\mathbf{X})$ is a fixed point**—not an isolated equilibrium but a whole set of equilibria.

5.3 Minimum Norm: $\hat{\beta}'$ Is the Point of the Solution Manifold Closest to the Origin

Theorem 5.1 (Minimum-norm solution). *On the solution manifold $\{\beta : \mathbf{X}\beta = \mathbf{y}\}$, $\hat{\beta}' = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{y}$ is the unique point of smallest Euclidean norm.*

证明. Take any solution β and write $\beta = \hat{\beta}' + \mathbf{w}$ with $\mathbf{X}\mathbf{w} = \mathbf{0}$ (the difference of two solutions lies in the null space). The key observation: $\hat{\beta}' \perp \mathcal{N}(\mathbf{X})$, because

$$\langle \hat{\beta}', \mathbf{w} \rangle = \mathbf{y}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X} \mathbf{w} = 0.$$

Pythagoras then gives

$$\|\beta\|^2 = \|\hat{\beta}'\|^2 + \|\mathbf{w}\|^2 \geq \|\hat{\beta}'\|^2,$$

with equality if and only if $\mathbf{w} = \mathbf{0}$. So $\hat{\beta}'$ is exactly the point where the solution manifold meets $\text{row}(\mathbf{X})$ (the column space of $\mathbf{X}^\top$). $\square$

5.4 Stability: Contraction on the Row Space, Neutrality on the Null Space

The set of fixed points is large; does the iteration converge, and to what? Carry over the stability analysis of Chapter 4, changing only the spectrum. The GD map $G(\beta) = \beta - \eta \nabla L(\beta)$ —itself linear—satisfies

$$G(\beta) - G(\beta_0) = (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})(\beta - \beta_0).$$

The eigenvalues of $\mathbf{X}^\top \mathbf{X}$: n positive ones $\lambda_1 \geq \dots \geq \lambda_n > 0$ (identical to the nonzero eigenvalues of $\mathbf{X}\mathbf{X}^\top$), plus $d - n$ zeros. So the eigenvalues of $\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}$ are

$$\underbrace{1 - \eta \lambda_j, j = 1..n}_{n \text{ of them}}; \quad \underbrace{1}_{d-n \text{ of them}}.$$

- **Row-space directions** (the complement of the $d - n$ zero-eigenvalue directions): when $0 < \eta < \frac{2}{\lambda_{\max}(\mathbf{X}\mathbf{X}^\top)}$, $|1 - \eta \lambda_j| < 1$ —**contraction**, the same exponential decay as in the $n \gg d$ case;
- **Null-space directions**: the factor is identically 1 —**neutral**: the $\mathcal{N}(\mathbf{X})$ component of the initial error is **preserved as is, forever**.

The conclusions, stated in layers:

- (1) **Every point of the solution manifold $\hat{\beta}' + \mathcal{N}(\mathbf{X})$ is Lyapunov stable:** transverse (row-space) perturbations are exponentially pressed back toward the manifold, tangential (null-space) perturbations neither grow nor vanish, and the iterate stays in a bounded neighborhood of the perturbed point;
- (2) **As isolated points,** none of the equilibria on the manifold is asymptotically stable (a small tangential perturbation does not return) —what is stable is **the manifold itself:** it is normally attracting;
- (3) **The initialization decides the landing point:** since the null-space component is conserved, the null-space component of $\hat{\beta}_0$ is carried all the way to the limit —

$$\lim_{t \rightarrow \infty} \hat{\beta}_t = \hat{\beta}' + \underbrace{P_{\mathcal{N}(\mathbf{X})} \hat{\beta}_0}_{\text{null-space component}},$$

where $P_{\mathcal{N}(\mathbf{X})}$ is the orthogonal projection onto the null space.

5.5 The Exact Solution, Continued: GD from Zero = Implicit Minimum-Norm Regularization

Take the most common initialization $\hat{\beta}_0 = \mathbf{0}$: its null-space component is zero, and the iteration **stays in row(X) for its entire run** (inductively, $\hat{\beta}_t = \mathbf{X}^\top \mathbf{u}_t$ for some $\mathbf{u}_t$, since the gradient $\mathbf{X}^\top(\cdot)$ always pulls vectors back into the row space). There is only one interpolation point in the row space — $\hat{\beta}'$ itself. Hence:

Theorem 5.2 (Exact solution for GD in the underdetermined case). *Let $\mathbf{X}$ have full row rank, $0 < \eta < 2/\lambda_{\max}(\mathbf{X}\mathbf{X}^\top)$, and $\hat{\beta}_0 = \mathbf{0}$. Then*

$$\hat{\beta}_t = \left[\mathbf{I} - (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t \right] \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{y} = \left[\mathbf{I} - (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t \right] \hat{\beta}' \quad (27)$$

and $\hat{\beta}_t \rightarrow \hat{\beta}'$ (the minimum-norm interpolating solution), with convergence rate governed by the condition number of $\mathbf{X}\mathbf{X}^\top$: $\|\hat{\beta}_t - \hat{\beta}'\| = O\left(\left(\frac{\kappa-1}{\kappa+1}\right)^t\right)$ (at the optimal step size).

证明. Use the identity $\mathbf{X}^\top (\mathbf{I} - \eta \mathbf{X}\mathbf{X}^\top) = (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}) \mathbf{X}^\top$ (both sides equal $\mathbf{X}^\top - \eta \mathbf{X}^\top \mathbf{X}\mathbf{X}^\top$) to unroll (18), obtaining $\hat{\beta}_t = \mathbf{X}^\top [\eta \sum_{k=0}^{t-1} (\mathbf{I} - \eta \mathbf{X}\mathbf{X}^\top)^k] \mathbf{y}$. The bracket is a geometric series of $n \times n$ matrices, and $\eta < 2/\lambda_{\max}$ keeps its spectral radius below 1 (strict contraction of all eigenvalues), so $\eta \sum_{k=0}^{t-1} (\mathbf{I} - \eta \mathbf{X}\mathbf{X}^\top)^k = [\mathbf{I} - (\mathbf{I} - \eta \mathbf{X}\mathbf{X}^\top)^t] (\mathbf{X}\mathbf{X}^\top)^{-1}$; substituting back and reusing the identity gives (27). The limit follows from $(\mathbf{I} - \eta \mathbf{X}\mathbf{X}^\top)^t \rightarrow \mathbf{0}$; the rate is Corollary 4.2 verbatim, with the spectrum of $\mathbf{X}^\top \mathbf{X}$ replaced by that of $\mathbf{X}\mathbf{X}^\top$ (same nonzero eigenvalues). $\square$

Place (27) next to (22): the structures are identical —**the only difference is the “target”**: $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is replaced by $\hat{\beta}' = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{y}$. The formal symmetry is no coincidence; the next section names the object that governs both.

Remark (Implicit regularization). Theorem 5.2 is the prototype of the modern theory of *implicit regularization* / *implicit bias*: **with no explicit penalty whatsoever, plain gradient descent from the origin automatically selects, among all interpolating solutions, the one of smallest norm—which is often the one that generalizes best.** The phenomenon that overparameterized neural networks “memorize without exploding and still generalize” has, in this linear version, its cleanest mathematical embodiment.

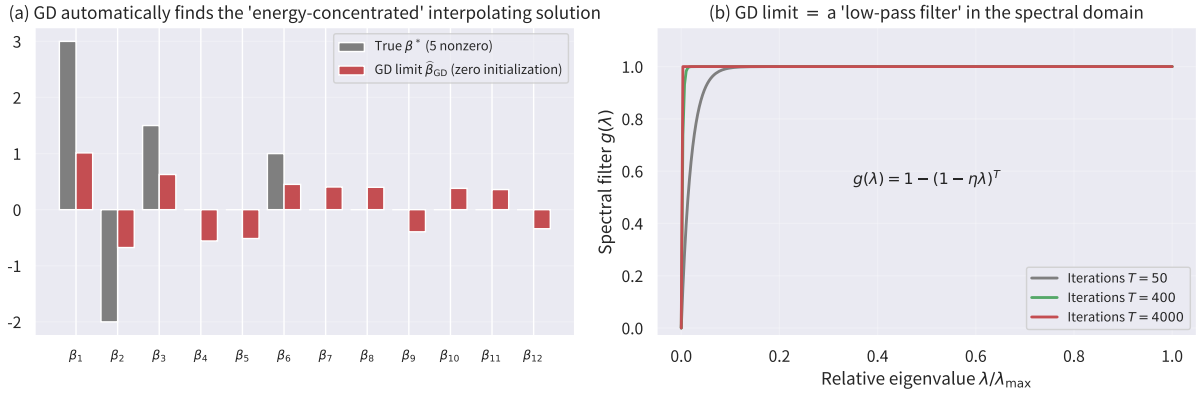


图 7: Two faces of implicit regularization ($d = 100 > n = 30$, noiseless interpolation): (a) the limit of zero-initialized GD (red) recovers energy along the true signal directions (gray, 5 nonzero coefficients) while automatically shrinking along the noise directions; (b) the spectral-filtering interpretation — t steps of GD apply the factor $g(\lambda) = 1 - (1 - \eta\lambda)^t$ along each eigen-direction λ : large-eigenvalue directions converge first, small-eigenvalue directions last, equivalent to a “low-pass filter.”

5.6 Comparison with the Pseudoinverse: One Object, Several Faces

Now for the comparison this chapter owes you. The Moore–Penrose pseudoinverse $\mathbf{X}^+$ is defined uniformly via the SVD: if $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ ($\mathbf{\Sigma}$ diagonal with singular values $s_j > 0$), then

$$\mathbf{X}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top, \quad \mathbf{\Sigma}^+ \text{ replaces each } s_j \text{ by } 1/s_j \text{ (zeros stay zero)}.$$

It yields the **minimum-norm least-squares solution** $\mathbf{X}^+\mathbf{y}$ for **any** $\mathbf{X}$. In the two full-rank regimes it collapses to the two familiar formulas:

- **Full column rank** ($n \gg d$): $\mathbf{X}^+ = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ —the OLS closed form of Section 3.2;
- **Full row rank** ($d > n$): $\mathbf{X}^+ = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}$ —**exactly this chapter’s $\hat{\beta}'$** . (Verify by SVD: $\mathbf{X}\mathbf{X}^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^\top\mathbf{U}^\top$, whose inverse is $\mathbf{U}(\mathbf{\Sigma}\mathbf{\Sigma}^\top)^{-1}\mathbf{U}^\top$; left-multiplying by $\mathbf{X}^\top = \mathbf{V}\mathbf{\Sigma}^\top\mathbf{U}^\top$ gives $\mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top$.)

So: **in the full-row-rank case $d > n$, $\hat{\beta}'$ is the pseudoinverse solution** —not “another solution,” but the same object seen in the underdetermined regime. The table below gathers every route, appearing throughout the book, to “the same point”:

Object	Formula	Applies when	How it is reached
OLS closed form	$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$	$n \gg d$, full column rank	Solve the normal equations directly
This chapter's $\hat{\beta}'$	$\mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y}$	$d > n$, full row rank	Solve the interpolation equations directly
Pseudoinverse solution $\mathbf{X}^+ \mathbf{y}$	SVD definition	any rank	The uniform definition (parent of the two above)
GD, $\hat{\beta}_0 = \mathbf{0}$	$\lim_t \hat{\beta}_t$	any rank	Implicit convergence to $\mathbf{X}^+ \mathbf{y}$ (exact solutions of these two chapters)
ridge, $\lambda \rightarrow 0^+$	$\lim_{\lambda \rightarrow 0^+} (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$	any rank	Equals $\mathbf{X}^+ \mathbf{y}$ (limit of an identity; see exercises)

Two comparison conclusions deserve separate remembering:

- (1) **Algebraically:** $\hat{\beta}' = \mathbf{X}^+ \mathbf{y}$ (full row rank) —it coincides with the pseudoinverse solution **exactly**, including the “minimum norm” property: among all defining properties of the pseudoinverse, the condition $\mathbf{X}^+ \mathbf{y} \perp \mathcal{N}(\mathbf{X})$ is Theorem 5.1 in the underdetermined case.
- (2) **Dynamically:** the pseudoinverse is “static” (the SVD carves out the answer in one cut), GD is “dynamic” (a polynomial inches toward it): (27) says the t -step GD solution equals $[\mathbf{I} - (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t] \mathbf{X}^+ \mathbf{y}$, the bracket a contraction polynomial acting on the pseudoinverse solution. Change the initialization and GD lands at **other** interpolating points (carrying the initialization’s null-space component) —*the GD family reaches the entire solution manifold; the pseudoinverse (and the zero initialization) selects the point of the manifold closest to the origin.*

Chapter takeaways

- When $d > n$ with full row rank, $\hat{\beta}' = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y}$ satisfies (i) exact interpolation $\mathbf{X} \hat{\beta}' = \mathbf{y}$ and (ii) $\nabla L(\hat{\beta}') = \mathbf{0}$ (first-order equilibrium), and is the **unique minimum-norm** point of the solution manifold;
- The entire solution manifold $\hat{\beta}' + \mathcal{N}(\mathbf{X})$ consists of GD fixed points: row-space directions contract exponentially (for $0 < \eta < 2/\lambda_{\max}(\mathbf{X} \mathbf{X}^\top)$), null-space directions are neutral —the manifold is transversally attracting, its points only Lyapunov stable, and the landing point is decided by the initialization’s null-space component;
- Exact solution $\hat{\beta}_t = [\mathbf{I} - (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t] \hat{\beta}'$: structurally isomorphic to the $n \gg d$ case, with the target switched from $\hat{\beta}$ to $\hat{\beta}'$; for $\hat{\beta}_0 = \mathbf{0}$, GD converges **implicitly** to the minimum-norm solution —the linear prototype of implicit regularization;
- Relation to the pseudoinverse: with full row rank, $\hat{\beta}' = \mathbf{X}^+ \mathbf{y}$ identically; the pseudoinverse, the two closed forms, the GD-from-zero limit, and the ridge zero limit are **five faces of one minimum-norm least-squares solution**.

Exercise 5.1. Prove the identity $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I})^{-1}$ (multiply both sides by $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})$ and rearrange), and use it to show that the ridge solution converges to $\mathbf{X}^+ \mathbf{y}$ as $\lambda \rightarrow 0^+$ (no full-rank assumption needed).

Exercise 5.2. Let $d > n$ with $\mathbf{X}$ of full row rank. Show that for any initialization $\hat{\beta}_0$, the GD limit is $\hat{\beta}' + P_{\mathcal{N}(\mathbf{X})} \hat{\beta}_0$, where $P_{\mathcal{N}(\mathbf{X})} = \mathbf{I} - \mathbf{X}^+ \mathbf{X}$ is the orthogonal projection onto the null space; and explain why this is consistent with Theorem 5.2.

Exercise 5.3. Compare the “effective dimension” of the two regimes: for $n \gg d$, the GD error decay is governed by the d eigenvalues of $\mathbf{X}^\top \mathbf{X}$; for $d > n$, by the n eigenvalues of $\mathbf{X}\mathbf{X}^\top$. Prove that the two matrices have exactly the same nonzero eigenvalues (hint: if $\mathbf{X}^\top \mathbf{X}\mathbf{v} = \lambda\mathbf{v}$, look at $\mathbf{X}\mathbf{v}$), and explain why this spectral identity lets the convergence-rate formulas of the two chapters be interchanged.

6 From Estimation to Inference: Covariance, Precision, and Conditional Independence

Chapter 2 settled “correlation $\neq$ causation” in one dimension. This chapter upgrades the question to the joint level: treating $(y, \mathbf{x})$ as a single $(d+1)$ -dimensional random vector, we ask how the **covariance matrix and its inverse (the precision matrix)** encode the correlation, conditional independence, and —under graph assumptions—causal structure between y and $\mathbf{X}$; and then we answer the practical questions: how to estimate when d is large and n is small (lasso, elastic net, CLIME, graphical lasso), and how to upgrade “point estimation” into “inference” (the debiased lasso) —in short, **from estimate to inference**.

6.1 Covariance and Precision: Two Encodings of Correlation

Let $(y, \mathbf{x})$ be jointly Gaussian with mean zero (w.l.o.g.), and write

$$\Sigma = \text{Cov} \begin{pmatrix} y \\ \mathbf{x} \end{pmatrix} \succ \mathbf{0}, \quad \Theta = \Sigma^{-1} \text{ (the precision matrix).}$$

The two matrices encode two different kinds of “correlation”:

- $\Sigma_{ij} \neq 0$: marginal correlation. It *propagates*: even when $y \perp x_j$ given the remaining variables (no direct link), x_j can remain marginally correlated with y —as long as an indirect path through other variables exists (the multivariate version of the common-cause structure of Section 2.5).
- $\Theta_{ij} = 0$: **conditional independence**. In the Gaussian case there is an exact identity

$$\Theta_{ij} = 0 \iff i \perp j \mid \text{all remaining variables}, \quad \rho_{ij \cdot V \setminus \{i,j\}} = -\frac{\Theta_{ij}}{\sqrt{\Theta_{ii} \Theta_{jj}}}. \quad (28)$$

The right-hand side is the *partial correlation*: *zero partial correlation $\Leftrightarrow$ independence given everything else*. The proof is just block inversion (Exercise 6.1).

(28) is the hinge of this chapter: **the zero pattern of the precision matrix is the adjacency matrix of a “conditional-independence graph”** —the Gaussian graphical model[15]. Correlation *flows* along the graph (indirect paths create correlation); conditional independence holds exactly where there is *no edge*. This graph is the Gaussian incarnation of Wright’s path diagram (Section 2.5).

6.2 Regression Coefficients Are Entries of the Precision Matrix

Now plug linear regression into this picture. Block the joint Gaussian precision as $\Theta = \begin{pmatrix} \Theta_{yy} & \Theta_{yX} \\ \Theta_{Xy} & \Theta_{XX} \end{pmatrix}$ (scalar $\Theta_{yy} > 0$, $\Theta_{Xy} \in \mathbb{R}^d$). By block inversion (Exercise 6.1), the conditional distribution of y given $\mathbf{X}$ is

$$y \mid \mathbf{X} \sim \mathcal{N} \left(-\frac{1}{\Theta_{yy}} \Theta_{yX}^\top \mathbf{x}, \frac{1}{\Theta_{yy}} \right). \quad (29)$$

Comparing with the regression model $y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon$ yields a set of identities that deserve to be carved in stone:

$$\boxed{\boldsymbol{\beta} = -\frac{\boldsymbol{\Theta}_{Xy}}{\boldsymbol{\Theta}_{yy}}, \quad \text{Var}(\varepsilon) = \frac{1}{\boldsymbol{\Theta}_{yy}}} \quad (30)$$

Regression = probing the graph

(30) says: **regression coefficients (together with their signs and significance) are ratios of precision-matrix entries.** In particular,

$$\beta_j = 0 \iff \Theta_{Xy,j} = 0 \iff y \perp x_j \mid \mathbf{x}_{-j}.$$

i.e., y and x_j are independent given all the other features.

This sentence translates one statistical problem into two:

- (1) **Correlation** (Chapter 2): are y and x_j marginally correlated? —read off $\boldsymbol{\Sigma}$, one t -test;
- (2) **Conditional independence**: controlling for everything else, is there still a “direct” link between y and x_j ? —read off the zero pattern of $\boldsymbol{\Theta}$, i.e. whether the **partial correlation** is zero.

The gap between the two is the matrix language of phenomena like “the correlation between x_j and y is entirely explained by a third variable” (confounding, mediation). Note that $\beta_j = 0$ delivers a **graph-theoretic “no direct edge”**, not a causal claim —unless the graph itself has already been pinned down as identifiable by external knowledge (Section 2.5) or by the conditional-independence tests below.

6.3 Conditional-Independence Testing and Causality: From Correlation to Graph

(28) supplies **testable constraints**: every missing edge on a Gaussian graph corresponds to a statistical hypothesis of zero partial correlation

$$H_0^{(ij)} : \rho_{ij \cdot V \setminus \{i,j\}} = 0 \quad (\iff \Theta_{ij} = 0), \quad (31)$$

which Fisher’s z -transform can test directly in low dimensions; in high dimensions individual testing becomes infeasible, but the logic of the constraints is unchanged. The PC algorithm (Spirtes, Glymour & Scheines) shows how such constraints turn into causal discovery[16]:

- (a) **Edge deletion**: for each pair of variables, search over conditioning sets S ; as soon as $i \perp j \mid S$ is accepted, delete the undirected edge between i and j —common causes and indirect paths are exposed by exactly this test (the operable version of Reichenbach’s principle[7]);
- (b) **Orientation**: for structures $i - k - j$ with i, j nonadjacent (v-structures), Markov + faithfulness allow orienting $i \rightarrow k \leftarrow j$ —**the data itself gives the direction of some arrows**;

(c) **Propagation:** Meek’s rules propagate orientations further along the graph.

Echoing Chapter 2: the correlation coefficient never yields direction, but **a battery of conditional-independence tests** can recover part of the DAG under the additional assumption that the distribution is faithful to a graph; the validity of causal conclusions still comes from the assumption list, not from the data itself —except that now each assumption can be tested and refuted one by one.

6.4 Sparsity: Why “Most Coefficients Are Zero” Is a Computable Assumption

The preceding sections gave the language of graphical models: up to $\binom{d}{2}$ edges among d variables, d^2 precision-matrix entries to estimate. When $d > n$ this is hopeless —unless we are willing to believe something very specific: **the true graph is sparse**, and most $\Theta_{ij} = 0$. This section explains why the assumption is not only plausible but “computable.”

Sparsity is a falsifiable assumption, not a prayer. “Most coefficients are zero” means the graph has $s \ll d^2$ edges; when the true edge count satisfies $s \log d \ll n$, a family of methods (next section) recovers the graph structure **exactly** with high probability. Conversely, if the world is truly dense, sparse methods fail honestly —cross-validation will not settle on a stable structure. This matches the Chapter 2.5 spirit of “fit first, then audit the assumptions”: the sparsity assumption can and should be audited.

Where does sparsity come from? Three independent sources. (1) **Scientific:** in gene regulatory networks each gene is regulated by few transcription factors; in social networks each person has few close contacts —“bounded nonzeros per row” is prior knowledge in many fields, mathematically $s = O(d)$ edges. (2) **Statistical:** under suitable designs, a sparse model’s effective degrees of freedom are $\ll d$ (below), and the sample complexity is governed by the **number of nonzeros** rather than the dimension —this is the very meaning of the term “high-dimensional statistics”: d may exceed n as long as the support of the true parameter is smaller than n . (3) **Computational:** sparsity turns a d^2 -dimensional continuous optimization into “find which coordinates are nonzero + estimate their values,” and the latter has explicit algorithms and error bounds.

Three faces of sparsity, one sentence each.

- (1) **Combinatorial face:** model selection —finding the correct one among 2^d subsets. The general problem is NP-hard, but under sparsity it becomes tractable (small s collapses the search space).
- (2) **Geometric face:** the ℓ_0 ball $\{\beta : \|\beta\|_0 \leq s\}$ is nonconvex, and ℓ_1 is its **convex relaxation** (the sharp envelope relation is $\|\beta\|_1 \leq \sqrt{s} \|\beta\|_2$; “the convex hull is exactly the ℓ_1 ball” holds only for $s = 1$): the relaxed problem is convex again, and the price of relaxation (recovery conditions) can be quantified via incoherence and the restricted eigenvalue condition.

- (3) **Statistical face**: effective degrees of freedom — the lasso fit $\mathbf{X}\hat{\boldsymbol{\beta}}$ has $\hat{df} = \mathbb{E}[\|\hat{\boldsymbol{\beta}}\|_0]$, so a **sparse model’s complexity is charged by the variables it actually uses**, not by d . This explains why the variance of a sparse estimator can be far below $O(d/n)$.

An honest caveat: sparsity is an **assumption about the world**, not a mathematical trick. If the true signal is dense but decaying (e.g. $\beta_j \propto 1/j$), strongly sparse methods will systematically miss weak signals —ridge (dense shrinkage) may then be the more honest choice. Choosing lasso over ridge is choosing which world you believe; the bias–variance language of Section 3.6 quantifies the trade-off.

With these three faces in hand, the next section turns to concrete methods of estimation: lasso, elastic net, graphical lasso, and CLIME —four methods, all nailing the assumption “sparse” into an estimator through different optimization forms.

6.5 Estimation: Four Regularizations for Large d , Small n

The sample covariance $\mathbf{S} = \frac{1}{n}\mathbf{X}^\top\mathbf{X}$ (after centering) is singular when $d > n$; $\mathbf{S}^{-1}$ does not exist —**without regularization there is no precision matrix**. Four approaches, one goal (sparse $\boldsymbol{\Theta}$):

The first two act directly on regression coefficients. The lasso (Tibshirani, 1996) solves

$$\min_{\boldsymbol{\beta}} \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1, \quad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^d |\beta_j|, \quad (32)$$

whose ℓ_1 geometry (the corners of the constraint set) drives some coordinates **exactly** to 0, yielding sparse solutions. Elastic net (Zou & Hastie, 2005) combines ℓ_1 with ℓ_2 :

$$\min_{\boldsymbol{\beta}} \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \left(\alpha \|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2} \|\boldsymbol{\beta}\|^2 \right), \quad (33)$$

where the ℓ_2 term lets **whole groups of highly correlated variables enter the model together** (fixing lasso’s habit of picking one member of a correlated group at random) —well suited to group-correlated data such as gene expression. The remaining two move the penalty to the matrix level:

- **Nodewise lasso** (Meinshausen & Bühlmann, 2006): regress each variable on all the others via (32), and connect i to j in the graph whenever a coefficient is nonzero —Section 6.2 says this is equivalent to recovering the precision matrix column by column [10, 17].
- **Graphical lasso** (Friedman, Hastie & Tibshirani, 2008): penalize the likelihood of the precision matrix directly,

$$\min_{\boldsymbol{\Theta} \succ \mathbf{0}} \text{tr}(\mathbf{S}\boldsymbol{\Theta}) - \log \det \boldsymbol{\Theta} + \lambda \|\boldsymbol{\Theta}\|_1,$$

estimating the whole graph in one shot; its optimality condition $\mathbf{S} - \boldsymbol{\Theta}^{-1} + \lambda \boldsymbol{\Gamma} = \mathbf{0}$ ($\boldsymbol{\Gamma}$ a subgradient) hooks estimation directly to conditional-independence testing [19].

- **CLIME** (Cai, Liu & Luo, 2011): leave the likelihood alone and solve

$$\min_{\Theta} \|\Theta\|_1 \quad \text{s.t.} \quad \|\mathbf{S}\Theta - \mathbf{I}\|_{\max} \leq \lambda,$$

i.e. “find a sparse matrix that is approximately the inverse of $\mathbf{S}$.” It is a **linear program** (column by column), needs no $\Theta \succ \mathbf{0}$ constraint and no log det computation, gives spectral-norm convergence rates under weaker assumptions, and natively fits scenarios where “ Σ need not be sparse, only Θ is”[20].

What the four share is the bias–variance trade (Section 3.6) replayed at the matrix level: λ crushes masses of Θ_{ij} to 0, trading a little bias for a controlled variance of the covariance inversion. Figure 8 plots the regularization path of (32) —its solution as a function of λ .

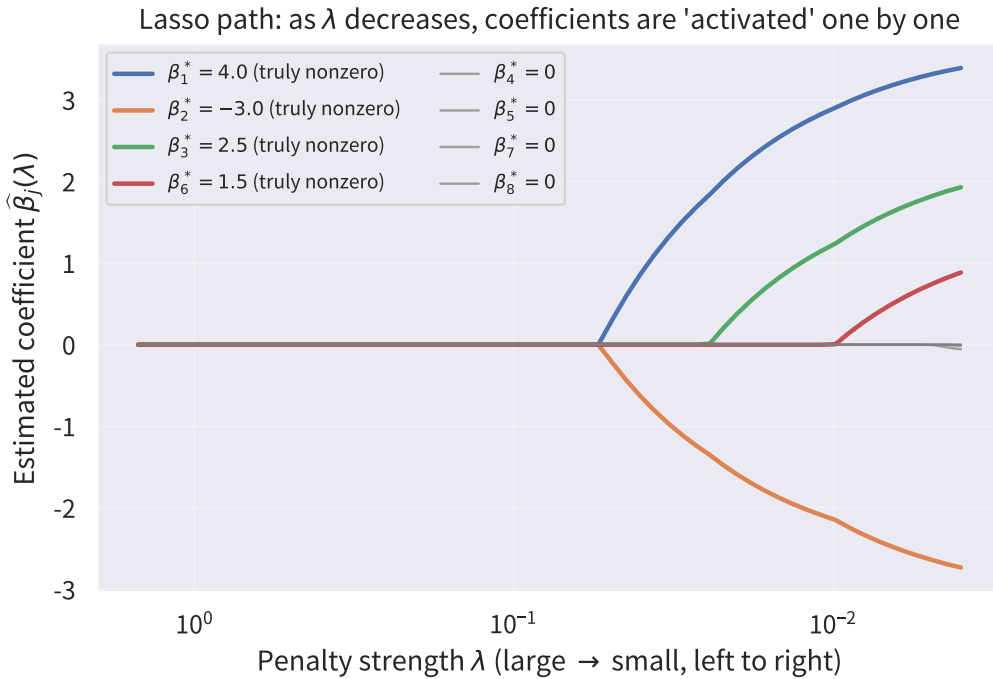


图 8: The lasso regularization path (simulated data, true coefficients 4, −3, 2.5, 0, 0, 1.5, 0, 0): as λ decreases (horizontal axis, large on the left), the four truly nonzero coefficients (colored) are “activated” one by one from 0, while the truly zero coefficients (gray) stay pinned at 0 —sparsity is not assumed but selected automatically by the geometry (diamond-shaped constraint corners) of the ℓ_1 penalty.

6.6 Inference: The Debiased Lasso and Tests of Conditional Independence

Once the estimate $\hat{\beta}_j$ is in hand, a **point estimate cannot be turned into a p -value directly**: the lasso’s shrinkage biases $\hat{\beta}_j$, and its limiting distribution is not Gaussian. The way out is **debiasing** (debiasing / desparsifying). The idea in one sentence: *add back what the penalty stole*. Take a nodewise estimate $\hat{\Theta}$ of the precision matrix (e.g. nodewise lasso regressions

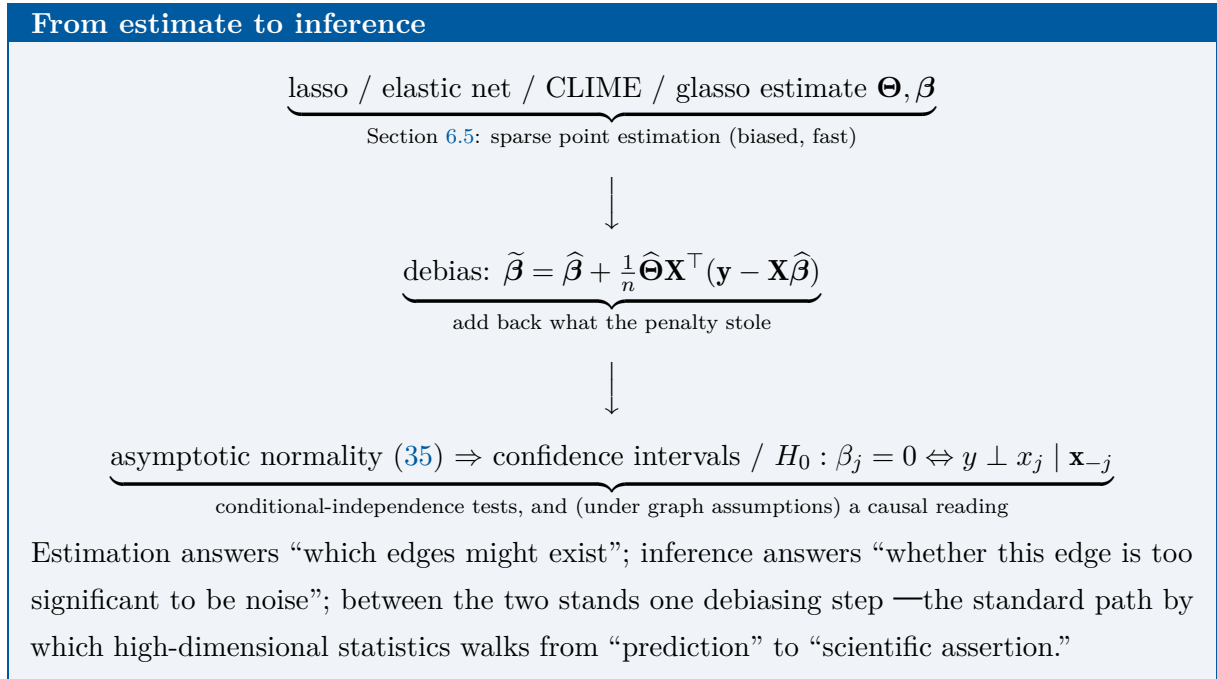
of $\mathbf{S}$, or CLIME directly), and define

$$\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}_{\text{lasso}} + \frac{1}{n} \hat{\boldsymbol{\Theta}} \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_{\text{lasso}}). \quad (34)$$

Algebraically (Exercise 6.2): $\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta} = \frac{1}{n} \hat{\boldsymbol{\Theta}} \mathbf{X}^\top \boldsymbol{\varepsilon} + (\mathbf{I} - \hat{\boldsymbol{\Theta}} \mathbf{S})(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_{\text{lasso}})$ —the first term is noise (asymptotically normal), the second a residual bias ($o_p(n^{-1/2})$ under sparsity). Hence (Zhang & Zhang, 2014; van de Geer et al., 2014)

$$\sqrt{n}(\tilde{\beta}_j - \beta_j) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2(\boldsymbol{\Theta} \boldsymbol{\Sigma} \boldsymbol{\Theta})_{jj}\right), \quad \text{confidence intervals: } \tilde{\beta}_j \pm z_{1-\alpha/2} \frac{\hat{\sigma}}{\sqrt{n}} \sqrt{(\hat{\boldsymbol{\Theta}} \mathbf{S} \hat{\boldsymbol{\Theta}})_{jj}}. \quad (35)$$

[21, 22] In high dimensions, **asymptotically normal inference for a single coefficient** is thus restored —and by (30), testing $H_0 : \beta_j = 0$ is **testing the conditional independence of y and x_j** . The whole pipeline closes:



Single-coefficient p -values aside, the working applied scientist faces an entire page of simultaneously tested conclusions —“how many of these are real” becomes the new question. Multiple-testing control of the false discovery rate (the BH procedure, knockoffs) is the subject of Chapter 7, and the closing of the loop of the sparsity story on the testing side.

Exercise 6.1. Block inversion: for $\boldsymbol{\Sigma} = \begin{pmatrix} \Sigma_{yy} & \Sigma_{yX} \\ \Sigma_{Xy} & \Sigma_{XX} \end{pmatrix}$ with inverse $\boldsymbol{\Theta} = \boldsymbol{\Sigma}^{-1}$, verify (29) and (30) from the blocks of $\boldsymbol{\Theta}$, and derive the partial-correlation formula (28).

Exercise 6.2. Derive the error decomposition $\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta} = \frac{1}{n} \hat{\boldsymbol{\Theta}} \mathbf{X}^\top \boldsymbol{\varepsilon} + (\mathbf{I} - \hat{\boldsymbol{\Theta}} \mathbf{S})(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_{\text{lasso}})$ for (34), and explain why the two terms are, respectively, “variance” and “bias.”

Exercise 6.3. Compare the optimality conditions of graphical lasso and CLIME: the former writes “ $\boldsymbol{\Theta}$ is the inverse of $\mathbf{S}$ ” as $\mathbf{S} - \boldsymbol{\Theta}^{-1} + \lambda \boldsymbol{\Gamma} = \mathbf{0}$ (depending on $\boldsymbol{\Theta}$ being positive definite and invertible), the latter as $\mathbf{S} \boldsymbol{\Theta} - \mathbf{I} \approx \mathbf{0}$ (linear constraints). Discuss: when $d \gg n$ and $\mathbf{S}$ is severely

ill-conditioned, which optimization problem is better posed? What practical advantages does CLIME's column-by-column LP structure bring?

7 Inference at Scale: False Discovery Rate Control

Section 6.6 upgraded a single coefficient β_j from a point estimate to a p -value. But the applied scientist never tests just one hypothesis: a single genomic screen tests $d \approx 2 \times 10^4$ genes at once, a marketing review inspects hundreds of segmented metrics, an epidemiological analysis regresses on dozens of exposures. **Chapter 6 answered “is *this* coefficient significant?”; this chapter answers “of this entire page of conclusions, how many are real”** —the last mile before regression output becomes a **credible scientific assertion**, and the closing of the loop from sparse estimation to FDR control.

7.1 Where the Problem Starts: Multiple Testing and the Credibility of Discoveries

Suppose we have m hypotheses $H_1, \dots, H_m$ (e.g. $H_j : \beta_j = 0$) with p -values $p_1, \dots, p_m$, of which m_0 are true. Testing each p -value at level α individually, if all m_0 null hypotheses were true, the probability of at least one false rejection is as high as

$$1 - (1 - \alpha)^{m_0} \approx 1 - e^{-\alpha m_0} \quad (\text{almost certain for large } m_0).$$

With $m = 10^4$ and $\alpha = 0.05$ this is essentially 100% ($1 - 0.95^{10^4} \approx 1 - e^{-513}$) —“**this page is entirely false discoveries**” is not an accident; it is the default outcome.

The classical remedy is to control the **family-wise error rate** (FWER): $\text{FWER} = \mathbb{P}(V \geq 1)$, where V is the number of false rejections. The Bonferroni method compresses each test’s threshold to α/m : $\text{FWER} \leq \alpha$ indeed holds, but at the price of a threshold that **tightens linearly in m** . At $m = 10^4$ the threshold is 5×10^{-6} : unless the effect is enormous, *nothing can be rejected* —the errors are controlled, and so are the discoveries.

In 1995 Benjamini and Hochberg proposed a question better matched to “screening” scenarios[31]: promise not “never make a single mistake” but control the **false discovery rate**

$$\text{FDR} = \mathbb{E} \left[\frac{V}{\max(R, 1)} \right], \quad R = \text{total number of rejections.} \quad (36)$$

V/R is **the fraction of false discoveries among the rejected** (with the convention 0 when $R = 0$). FDR allows false discoveries but requires them to be controlled *in proportion* —the more discoveries, the more false ones tolerated; and when nothing can be rejected, no debt is owed. FWER and FDR relate in one line (Exercise 7.2):

$$\text{FDR} \leq \text{FWER},$$

so Bonferroni, which controls FWER, necessarily controls FDR —just far too conservatively. The division of labor is therefore clear: **confirmatory** settings (one assertion going to court, onto a drug label) demand FWER; **exploratory** settings (screening candidate genes, shortlisting metrics) should use FDR.

7.2 The Benjamini–Hochberg Procedure: Ranking, a Sloped Line, and the Last Crossing Point

The BH procedure is disarmingly simple. Sort the p -values $p_{(1)} \leq \dots \leq p_{(m)}$; given a target level $q \in (0, 1)$, define

$$\hat{k} = \max \left\{ k : p_{(k)} \leq \frac{qk}{m} \right\}, \quad \text{reject } H_{(1)}, \dots, H_{(\hat{k})} \quad (\text{if no such } \hat{k} \text{ exists, reject nothing}). \quad (37)$$

Geometrically (Figure 9(a)): plot the sorted p -values as a staircase and find the **last crossing point** with the sloped line qk/m . Bonferroni is the horizontal line q/m (a single point at $k = 1$); BH relaxes the threshold *linearly* in the rank k —the k -th smallest p -value faces a hurdle k times Bonferroni’s.

Theorem 7.1 (Benjamini–Hochberg, 1995). *If the p -values are independent, or more generally satisfy a positive-regression-dependence condition (PRDS), then the BH procedure (37) satisfies*

$$\text{FDR} \leq \frac{m_0}{m} q \leq q.$$

The full proof under independence is the guided induction of Exercise 7.3; here is the intuition. The false-discovery proportion is adjudicated by the *slope* of the line: the relaxation rate of qk/m is calibrated so that per k rejections, at most qk false ones are budgeted. False discoveries enter the rejection set by “smuggling in as small p -values among the real signals”; and when real signals exist, $\hat{k} \gg 1$, so the threshold $q\hat{k}/m$ relaxes by orders of magnitude relative to q/m —**the more discoveries, the looser each individual hurdle, but the total budget is locked at q** . This is the precise mechanism of “tolerate a few false ones when there are many discoveries.”

7.3 Closing the Loop: BH Is an Adaptive Sparse Estimator

Now back to this book’s main line —sparsity. Transform one-sided p -values into z -scores $z_j = \Phi^{-1}(1 - p_j)$; the BH procedure is equivalent to finding a **data-driven cutoff** on the ordered z -scores: reject $z_{(j)} \geq \hat{t}$. This cutoff is not the pre-specified $\sqrt{2 \log m}$; it “grows” out of the data itself —and where it lands depends on how sparse the signals are.

Abramovich, Benjamini, Donoho, and Johnstone (2006) turned this observation into a theorem[32]. Consider the sparse normal means model

$$z_j = \theta_j + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, 1), \quad \#\{j : \theta_j \neq 0\} \leq s \ll m,$$

and use the BH cutoff at an appropriate level q as a *threshold estimator* $\hat{\theta}_j = z_j \cdot \mathbf{1}\{z_j \geq \hat{t}\}$ (hard thresholding). The core result: **without knowing the sparsity s , this FDR-controlled rule automatically achieves the same minimax risk rate as the oracle threshold that *knows* s** —squared risk on the order of $2s \log(m/s)$ (up to constant factors), rather than the $2s \log m$ delivered by the non-adaptive threshold $\sqrt{2 \log m}$. When $s \ll m$, $\log(m/s) \ll \log m$, and the gap is substantial.

Placed side by side with Section 6.5, the same face appears twice:

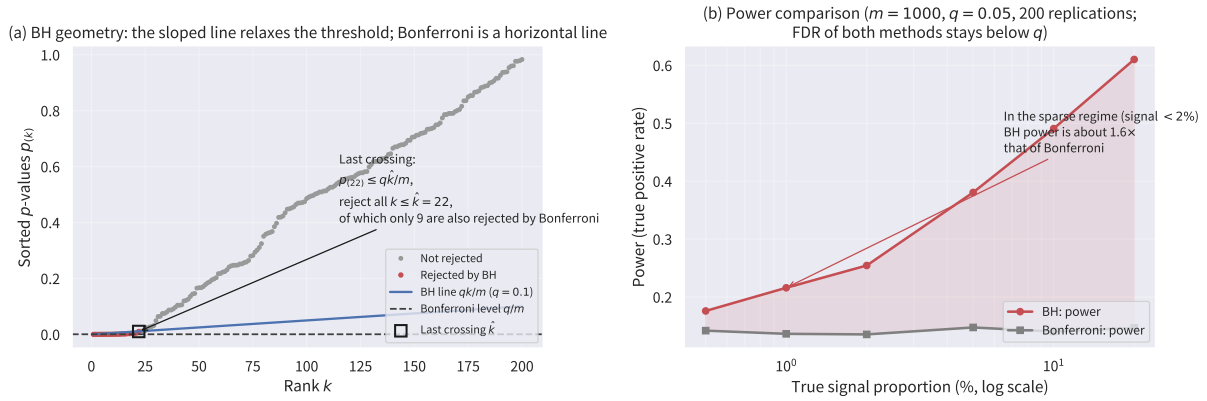


图 9: (a) The geometry of BH (one simulation: $m = 200$ independent tests, of which 30 signals with $z_j \sim \mathcal{N}(3.2, 1)$, $q = 0.1$): the staircase of sorted p -values crosses the sloped line qk/m for the last time at $\hat{k} = 22$, and all $k \leq 22$ are rejected; Bonferroni’s horizontal line q/m would reject only 9 of them. (b) A power simulation ($m = 1000, q = 0.05, 200$ replications): as the signal fraction varies from 0.5% to 20% ($z_j \sim \mathcal{N}(3, 1)$), the power of BH rises from 0.18 to 0.61 while Bonferroni’s stays pinned near 0.14; at a 1% signal fraction BH’s power is about $1.6 \times$ Bonferroni’s. Both procedures keep the measured FDR at or below q (BH between 0.038 and 0.054, Bonferroni between 0.0008 and 0.045) —**the power bought by the sloped line costs not a cent of error rate.**

	Sparse estimation (Chapter 6)	FDR testing (this section)
Object	regression coefficients β or means θ	a family of hypotheses $H_1, \dots, H_m$
Mechanism	ℓ_1 penalty / soft thresholding	sorted p -values / BH threshold
Source of adaptivity	penalty parameter λ via cross-validation or BIC	cutoff located automatically by the crossing point
Role of sparsity	makes estimation possible when $d \gg n$	lets the BH cutoff drop automatically to the signal layer

Controlling the false discovery rate is not a patch on top of estimation; it is the dual formulation of the sparsity assumption on the testing side: “most coefficients are zero” both enables the lasso to estimate and enables BH to adaptively find its threshold. Donoho’s name appears once in Chapter 6 (the compressed-sensing tradition) and once in this section — no coincidence: it is the same logic chain “sparsity $\Rightarrow$ thresholding $\Rightarrow$ optimality,” seen from its two ends.

7.4 Back to Regression I: Debiased Lasso p -Values Through BH Control the FDR of Coefficients

The pipeline is already in place: Section 6.6 gives the asymptotic normality of $\sqrt{n}(\tilde{\beta}_j - \beta_j)$ in (35), hence an asymptotic p -value p_j for every $H_{0j} : \beta_j = 0$. Feed the d p -values to BH (37); by Theorem 7.1 —provided the p -values are approximately independent or positively correlated —**the false discovery rate of the rejected coefficient set is controlled at q .** Combined with the correspondence $\beta_j = 0 \Leftrightarrow y \perp x_j \mid \mathbf{x}_{-j}$ of Section 6.2, this is the standard endpoint of conditional-independence structure screening in high-dimensional regression: lasso proposes

candidate edges, the debiased lasso produces p -values, BH certifies credibility.

Per this book’s custom, three honest caveats, disclosed one by one:

- (i) **The p -values are asymptotic.** (35) is a large-sample statement; in finite samples BH’s control is approximate, and the larger d is relative to n , the more the approximation leans on the sparsity conditions.
- (ii) **Correlation structure.** The debiased p -values are correlated through the design matrix. Extreme collinearity (the ill-conditioning of Section 3.6) can break the PRDS condition, and the measured FDR can overshoot; under general dependence there is the more conservative Benjamini–Yekutieli correction[33] (replace q by $q / \sum_{j=1}^m j^{-1} \approx q / \log m$), at a cost in power.
- (iii) **Consistency of screening and inference objects.** If lasso is used to pre-screen variables and tests are performed only on the survivors, standard BH theory no longer applies directly (the selection itself introduces extra randomness) —precisely the pitfall the method of the next section is designed to bypass.

7.5 Back to Regression II: Knockoffs —FDR Control Without p -Values

Barber and Candès (2015) took a more radical route[30]: **instead of manufacturing p -values and correcting them afterward, construct “mirror variables” inside the design matrix and let symmetry itself bear the control.** For each column $\mathbf{x}_j \in \mathbb{R}^n$, construct a knockoff $\tilde{\mathbf{x}}_j$ such that **swapping any pair $(\mathbf{x}_j, \tilde{\mathbf{x}}_j)$ leaves the joint distribution of the augmented design $[\mathbf{X} \ \tilde{\mathbf{X}}]$ unchanged** (the exchangeability property). For Gaussian designs there is an explicit construction —and it requires second-order information about the correlations among the columns of $\mathbf{X}$, i.e. **we are back to the covariance/precision matrices of Section 6.1**: each knockoff is calibrated to its original so that the more strongly correlated a variable, the more convincing its mirror. (A remark on conventions: the original Barber–Candès **fixed- X** construction requires $n \geq 2d$; the second-order construction described here is the **model- X** Gaussian version, whose high-dimensional construction remains an active research direction.)

Next define an importance score Z_j for each variable and $\tilde{Z}_j$ for its knockoff (e.g. the lasso-path order in which the variable enters the model), and combine them into the **signed statistic**

$$W_j = Z_j - \tilde{Z}_j,$$

whose sign records whether the original beats its mirror. The key observation: under H_{0j} ($\mathbf{x}_j$ is independent of $\mathbf{y}$), the original and the knockoff are indistinguishable, and exchangeability gives

$$W_j \stackrel{d}{=} -W_j \quad (H_{0j}, \text{ conditional on everything else}), \quad (38)$$

i.e. *the W_j of null variables are symmetric about 0*, while true signals push W_j toward large positive values. The count of *negative W ’s* therefore becomes a **symmetry-based estimate of**

the false discovery count: ordering by $|W_j|$ from largest to smallest, the knockoff+ procedure takes the threshold

$$T = \min \left\{ t > 0 : \frac{1 + \#\{j : W_j \leq -t\}}{\#\{j : W_j \geq t\}} \leq q \right\}, \quad \text{reject } \{j : W_j \geq T\}. \quad (39)$$

The “1+” in the numerator is an overdispersion correction for the randomness; and (38) guarantees that each falsely rejected variable contributes a negative mirror with probability 1/2 —the very fulcrum of the proof. The conclusion holds under *any* model (no linearity, no asymptotics):

$$\text{FDR} \leq q.$$

The two routes, side by side:

	BH + debiased lasso (Section 7.4)	Knockoffs (this section)
Fulcrum of control	asymptotic normality of p -values	exchangeability symmetry of the design
Model assumptions	required (linearity + sparsity + asymptotics)	none (model-free)
Finite-sample guarantee	approximate (asymptotic)	exact (finite-sample)
Extra cost	nodewise estimation needed for debiasing	constructing knockoffs (fixed- X construction needs $n \geq 2d$; model- X needs a known or estimable feature distribution)

At this point the “credible scientific assertion” toolkit is complete: the debiased lasso gives **single-coefficient** p -values (Section 6.6), BH gives FDR control for a **page** of p -values (earlier in this chapter), and knockoffs give model-free, finite-sample-exact FDR control for **variable selection** (this section). Candès and Donoho are honored in the preface because the modern credibility theory of regression happens to have been written by these two schools —one showing that *sparsity makes estimation possible*, the other that *symmetry makes inference credible*.

Chapter takeaways

- Under multiple testing, Bonferroni controls FWER but its α/m threshold tightens linearly in m and power collapses; $\text{FDR} = \mathbb{E}[V/\max(R, 1)]$ allows “tolerate a few false ones when there are many discoveries” —the right currency for exploratory screening (Section 7.1);
- The BH procedure = the last crossing point of sorted p -values with the sloped line qk/m ; under independence (or PRDS), $\text{FDR} \leq q m_0/m \leq q$ (Theorem 7.1);
- Closing the loop: the BH threshold is an adaptive sparse estimator of the z -scores, achieving oracle-level minimax rates without knowing the sparsity[32] —false discovery control and sparse estimation are one coin;
- The regression pipeline: debiased lasso emits p -values, BH controls FDR —but asymptotic approximation and collinearity are honestly disclosed weaknesses; knockoffs use the exchangeability symmetry of mirror variables to give model-free, finite-sample-exact FDR control[30].

Exercise 7.1. $m = 5$, $q = 0.1$, and five (already sorted) p -values 0.001, 0.02, 0.04, 0.09, 0.3. Execute BH step by step: write down the threshold qk/m for each k , the decision, $\hat{k}$, and the

final rejection set; then compare the rejection sets of BH, Bonferroni (threshold q/m), and naive per-test testing at level q .

Exercise 7.2. (a) Prove $\text{FDR} \leq \text{FWER}$, and conclude that any procedure controlling FWER (e.g. Bonferroni) automatically controls FDR; (b) prove that the Bonferroni rejection set at level q is a subset of the BH rejection set at level q ; (c) Let $m = 10^5$ independent tests include 200 signals with $z_j \sim \mathcal{N}(5, 1)$ (two-sided p -values), $q = 0.05$. Estimate Bonferroni's per-test threshold and power, quantify how much looser the BH threshold is at $\hat{k} \approx 200$, and explain why BH's power then tends to 1 while Bonferroni's falls well short.

Exercise 7.3 (Proof sketch, harder). Under **all m hypotheses true** and **independent p -values**, prove $\mathbb{P}(\exists k : p_{(k)} \leq qk/m) \leq q$ (under the complete null, FDR equals this probability, so the worst case of Theorem 7.1 holds). Complete the following skeleton: (i) induct on m , with $m = 1$ trivial; (ii) key step: condition on the largest order statistic $p_{(m)}$ and split into the cases $p_{(m)} \leq q$ and $p_{(m)} = u > q$; use the fact that, given $p_{(m)} = u$, the remaining $m - 1$ variables are $m - 1$ independent $U(0, u)$ variables divided by u , rewrite the event as the same event for $m - 1$ uniforms, and verify that the level parameter changes from q to $q' = q(m - 1)/(mu)$; (iii) combine the two cases.

Exercise 7.4. (a) Use exchangeability to show carefully: under H_{0j} , globally swapping $\mathbf{x}_j$ with $\tilde{\mathbf{x}}_j$ leaves the joint distribution of the statistic vector $(W_1, \dots, W_m)$ invariant while flipping the sign of W_j —hence (38) (hint: under H_{0j} , $\mathbf{y}$ does not depend on $\mathbf{x}_j$, so the swap does not change the regression problem); (b) explain why $\#\{j : W_j \geq t\}$ in the numerator of (39) counts “discoveries with $|W|$ above t ,” and why (up to the 1+) $\#\{j : W_j \leq -t\}$ is an **estimate of the false-discovery count** —and how symmetry turns this estimate into the fulcrum of an upper bound in expectation.

8 Nonlinear Latent Spaces: Linear Regression as a Natural Kernel Method

Chapter 1 said it: “linear” is a requirement on the **parameters**, not on the raw variables. This chapter pushes that sentence to its limit. Suppose the observed features are in fact nonlinear projections of high-dimensional latent variables

$$\mathbf{x}_i = \phi(\mathbf{z}_i), \quad \mathbf{z}_i \in \mathbb{R}^q \text{ (latent)}, \quad \phi : \mathbb{R}^q \rightarrow \mathbb{R}^d \text{ (feature map)}, \quad (40)$$

where **both the latent dimension q and the feature dimension d may be arbitrarily large** — including infinite: $\mathbf{z}$ itself may be a function (with ϕ mapping functions to functions), and as long as the kernel values $k(\mathbf{z}, \mathbf{z}')$ can still be evaluated, the entire machinery of this chapter runs unchanged. Running least squares on $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ is then simply pointing the entire machinery of Chapters 3–5 at the feature matrix

$$\Phi := [\phi(\mathbf{z}_1)^\top; \dots; \phi(\mathbf{z}_n)^\top] \in \mathbb{R}^{n \times d}.$$

The core conclusion of this chapter: **the linear regression between $\mathbf{X}$ and $\mathbf{y}$ behaves, in the world of the latent variables $\mathbf{z}$, as a kernel method — the prediction is nonlinear in $\mathbf{z}$, yet the computation only needs the $n \times n$ kernel matrix.**

8.1 Kernels and Gram matrices: the inner product as the only interface

Define the **kernel function** as the inner product in feature space

$$k(\mathbf{z}, \mathbf{z}') = \langle \phi(\mathbf{z}), \phi(\mathbf{z}') \rangle, \quad (41)$$

together with the sample Gram matrix and the kernel vector of a new point

$$\mathbf{K} = \Phi\Phi^\top \in \mathbb{R}^{n \times n}, \quad K_{ij} = k(\mathbf{z}_i, \mathbf{z}_j); \quad \mathbf{k}(\mathbf{z}) = \Phi\phi(\mathbf{z}) = (k(\mathbf{z}, \mathbf{z}_1), \dots, k(\mathbf{z}, \mathbf{z}_n))^\top. \quad (42)$$

Note that **no coordinate of ϕ appears in (42)** — the whole idea of kernel methods is: if every intermediate quantity of an algorithm can be written in terms of inner products $\langle \phi(\cdot), \phi(\cdot) \rangle$, then ϕ never needs to be constructed explicitly, and $d = \infty$ is just as fine. Mercer’s theorem guarantees that every positive-definite kernel k corresponds to some (possibly infinite-dimensional) feature map ϕ ; the two are two faces of the same object[12].

8.2 $d > n$: the kernel form of $\hat{\boldsymbol{\beta}}'$

Substitute Φ for $\mathbf{X}$ in the underdetermined closed form $\hat{\boldsymbol{\beta}}' = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{y}$ of Chapter 5. For a new sample $\tilde{\mathbf{x}} = \phi(\tilde{\mathbf{z}})$, the prediction is

$$\tilde{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}' = \phi(\tilde{\mathbf{z}})^\top \Phi^\top (\Phi\Phi^\top)^{-1} \mathbf{y} = \mathbf{k}(\tilde{\mathbf{z}})^\top \mathbf{K}^{-1} \mathbf{y}. \quad (43)$$

Check every object in this line step by step: $\Phi\Phi^\top = \mathbf{K}$ is the kernel matrix; $\phi(\tilde{\mathbf{z}})^\top\Phi^\top = \mathbf{k}(\tilde{\mathbf{z}})^\top$ is the kernel vector. **The prediction is written entirely in terms of the kernel function.** Written as an expansion,

$$\hat{f}(\tilde{\mathbf{z}}) = \tilde{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}' = \sum_{i=1}^n \alpha_i k(\tilde{\mathbf{z}}, \mathbf{z}_i), \quad \boldsymbol{\alpha} = \mathbf{K}^{-1} \mathbf{y}, \quad (44)$$

i.e., the prediction is a **linear combination of kernel sections anchored at the training points** — this is **kernel interpolation**, also known as kernel ridgeless regression: it is the minimum-norm interpolating solution in $d > n$ feature space (Section 5.3) expressed in kernel language, and for a positive-definite kernel the feature-space norm is exactly the RKHS norm — **the point GD converges to from zero is simultaneously the interpolating function of minimal norm in the RKHS**[13]. This is the kernel form of the $(\phi(\mathbf{Z})\phi(\mathbf{Z})^\top)^{-1}$ in the original question: *it never needs to be constructed — it is simply $\mathbf{K}^{-1}$, assembled directly from kernel values.*

8.3 $n \gg d$: the kernel form of $\hat{\boldsymbol{\beta}}$ and the push-through identity

Substitute Φ for $\mathbf{X}$ in the closed form $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ of Section 3.2; the prediction becomes $\phi(\tilde{\mathbf{z}})^\top(\Phi^\top\Phi)^{-1}\Phi^\top\mathbf{y}$. Trouble: the entries of $\Phi^\top\Phi \in \mathbb{R}^{d \times d}$ are $\sum_i \phi_a(\mathbf{z}_i)\phi_b(\mathbf{z}_i)$ — **cross terms of feature coordinates, not determined by the kernel matrix $\mathbf{K}$ alone.** This is exactly the dividing line of “when kernelization is possible”: quantities that depend only on pairwise inner products of the observed points can be kernelized; quantities that depend on the feature coordinate system itself cannot.

The way out is the ridge regularization already familiar from Section 3.6 — it both repairs ill-conditioning and, as it happens, completes the kernelization. With λ added, the prediction is $\phi(\tilde{\mathbf{z}})^\top(\Phi^\top\Phi + \lambda\mathbf{I}_d)^{-1}\Phi^\top\mathbf{y}$, and the **push-through identity** (verify by multiplying both sides on the left by $(\Phi^\top\Phi + \lambda\mathbf{I})$)

$$(\Phi^\top\Phi + \lambda\mathbf{I}_d)^{-1}\Phi^\top = \Phi^\top(\Phi\Phi^\top + \lambda\mathbf{I}_n)^{-1} = \Phi^\top(\mathbf{K} + \lambda\mathbf{I}_n)^{-1}, \quad (45)$$

has only kernel objects on the right-hand side. Hence **kernel ridge regression**

$$\hat{f}(\tilde{\mathbf{z}}) = \mathbf{k}(\tilde{\mathbf{z}})^\top(\mathbf{K} + \lambda\mathbf{I}_n)^{-1}\mathbf{y}. \quad (46)$$

This is the kernel form of the $(\phi(\mathbf{Z})^\top\phi(\mathbf{Z}))^{-1}$ from the original question: *it is first regularized by λ , then flipped through the sample space by push-through, becoming $(\mathbf{K} + \lambda\mathbf{I}_n)^{-1}$ — the $d \times d$ inverse in the original space is never computed.* Letting $\lambda \rightarrow 0^+$, (46) formally recovers (43): **the two regimes converge in kernel language.** Gaussian process regression is the Bayesian retelling of the same object (kernel = covariance function, posterior mean = (46))[12].

8.4 Examples: when a kernel is “natural”

- **Polynomial kernel** $k(\mathbf{z}, \mathbf{z}') = (\mathbf{z}^\top\mathbf{z}' + c)^m$: corresponds to an explicit finite-dimensional feature map (all monomials of degree $\leq m$), with $d = \binom{q+m}{m}$ — a finite-dimensional regression with $n \gg d$, where kernelization is merely a convenience.

- **Gaussian kernel** (RBF) $k(\mathbf{z}, \mathbf{z}') = \exp(-\|\mathbf{z} - \mathbf{z}'\|^2 / 2\sigma^2)$: the corresponding ϕ is **infinite-dimensional** and cannot be constructed explicitly — here kernelization is not a trick but the *only* way out. The $d > n$ story of (43) naturally lives here.
- **Random features** (Rahimi & Recht, 2007): going the other direction, approximate the Gaussian kernel back to a **finite** dimension via $\phi(\mathbf{z}) = \sqrt{\frac{2}{D}} (\cos(\boldsymbol{\omega}_1^\top \mathbf{z} + b_1), \dots)$ (random frequencies) and then run all the d -dimensional linear algebra of this chapter — as $d \rightarrow \infty$, random-feature regression converges to kernel regression[14]. This route welds “kernel methods” and “wide neural networks” together: random-feature regression is precisely the linear version of the function-space limit of a two-layer infinitely wide network.

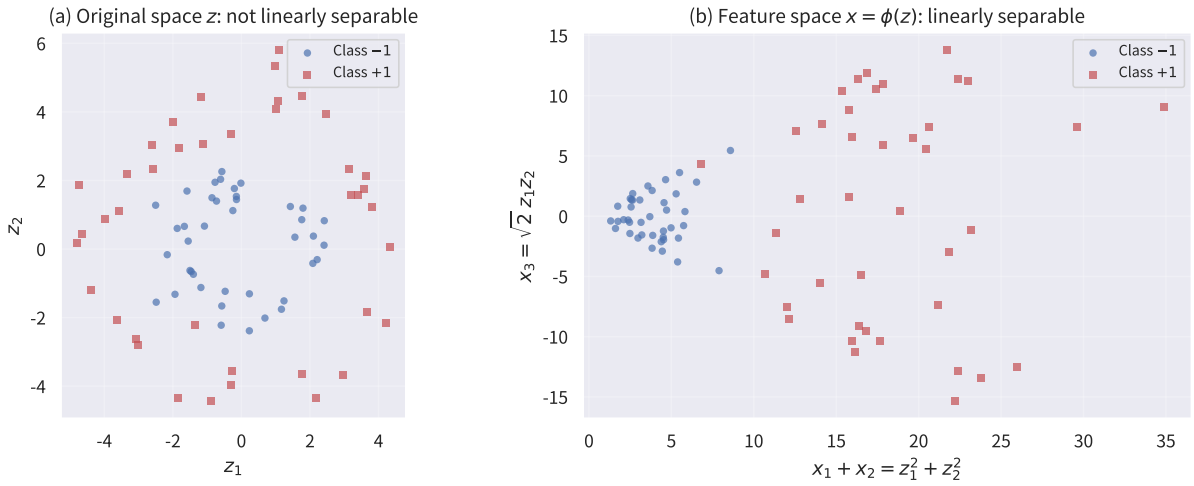


图 10: The geometric essence of kernel methods: the same ring-shaped data (two classes by radius) cannot be separated by any straight line in the original latent space $\mathbf{z}$ (left); lifted by the quadratic feature map $\phi(\mathbf{z}) = (z_1^2, z_2^2, \sqrt{2}z_1z_2)$ into feature space (right), a single linear function separates them. **Linear in x = nonlinear in $\mathbf{z}$** — and hence the prediction of linear regression becomes a natural nonlinear method in latent space.

8.5 From kernel regression to deep networks: ridgeless regression, Fourier features, and the neural tangent kernel

This chapter closes by doing something that “collects the whole book”: walking the road “linear regression $\rightarrow$ kernel regression” all the way to the doorstep of **deep neural networks**. There are two bridges along the way: the first is **random Fourier features** (the old bridge, 2007), the second is the **neural tangent kernel** (NTK, the new bridge, 2018)[24].

Bridge one: a two-layer network = linear regression on random features

Consider the simplest two-layer network

$$f(\mathbf{z}; \mathbf{a}) = \frac{1}{\sqrt{D}} \sum_{j=1}^D a_j \sigma(\boldsymbol{\omega}_j^\top \mathbf{z} + b_j), \quad (47)$$

where σ is an activation (ReLU, tanh, ...), the first-layer parameters (ω_j, b_j) are **randomly sampled from some distribution and then frozen**, and only the second layer $\mathbf{a} \in \mathbb{R}^D$ is trained. Note two things:

- (1) With $\phi(\mathbf{z}) = \frac{1}{\sqrt{D}}(\sigma(\omega_1^\top \mathbf{z} + b_1), \dots, \sigma(\omega_D^\top \mathbf{z} + b_D))^\top$, we have $f(\mathbf{z}; \mathbf{a}) = \phi(\mathbf{z})^\top \mathbf{a}$ — **it is simply linear regression in $\mathbf{a}$** , and the entire machinery of this chapter (normal equations, exact GD solution, push-through, the kernelization criterion) applies verbatim.
- (2) If $\mathbf{a}$ is initialized at zero and trained by gradient descent, the implicit-regularization analysis of Section 5.5 says: it converges to the **minimum-norm** interpolating solution — in the Euclidean norm of $\mathbf{a}$. This is **ridgeless regression** (ridge with $\lambda \rightarrow 0^+$, see Exercise 5.1): *with no explicit penalty at all, the interpolating solution of smallest norm is automatically selected by GD.*

By (43), its prediction takes the same kernel form, only with the kernel replaced by the **random-feature kernel**

$$k_{\text{RF}}(\mathbf{z}, \mathbf{z}') = \frac{1}{D} \sum_{j=1}^D \sigma(\omega_j^\top \mathbf{z} + b_j) \sigma(\omega_j^\top \mathbf{z}' + b_j) \xrightarrow{D \rightarrow \infty} \mathbb{E}_{\omega, b}[\sigma(\omega^\top \mathbf{z} + b) \sigma(\omega^\top \mathbf{z}' + b)], \quad (48)$$

which is exactly the theoretical core of Rahimi & Recht’s random Fourier features (for shift-invariant kernels, taking $\sigma = \cos$ with a specific sampling distribution makes (48) reproduce the Gaussian kernel exactly)[14]. **One sentence: freeze the first layer, train the second, initialize at zero, run gradient descent = ridgeless regression in a Fourier-feature space.**

Bridge two: the NTK — with all parameters trained, function space stays linear

Bridge one leaves an “unnatural” assumption: the first layer is frozen. What if **all** parameters $\boldsymbol{\theta} = (\mathbf{a}, \{\omega_j, b_j\})$ are trained? The loss in parameter space,

$$L(\boldsymbol{\theta}) = \frac{1}{2} \sum_{i=1}^n (f(\mathbf{z}_i; \boldsymbol{\theta}) - y_i)^2,$$

is in general **nonconvex** — the source of deep learning’s “mystery.” The key observation of Jacot, Gabriel, and Hongler is: stop staring at parameter space; stare at **function space**. For the residual vector $\mathbf{r}_t = (f(\mathbf{z}_i; \boldsymbol{\theta}_t) - y_i)_i$ on the training set, the chain rule gives

$$\frac{d f(\mathbf{z}; \boldsymbol{\theta}_t)}{dt} = \nabla_{\boldsymbol{\theta}} f(\mathbf{z}; \boldsymbol{\theta}_t)^\top \frac{d \boldsymbol{\theta}_t}{dt} = - \sum_{i=1}^n \underbrace{\langle \nabla_{\boldsymbol{\theta}} f(\mathbf{z}; \boldsymbol{\theta}_t), \nabla_{\boldsymbol{\theta}} f(\mathbf{z}_i; \boldsymbol{\theta}_t) \rangle}_{\Theta_t(\mathbf{z}, \mathbf{z}_i)} r_{t,i}, \quad (49)$$

where

$$\Theta_t(\mathbf{z}, \mathbf{z}') := \langle \nabla_{\boldsymbol{\theta}} f(\mathbf{z}; \boldsymbol{\theta}_t), \nabla_{\boldsymbol{\theta}} f(\mathbf{z}'; \boldsymbol{\theta}_t) \rangle \quad (50)$$

is the **neural tangent kernel**. The shape of (49) should look familiar: it is **kernel gradient descent in function space** — isomorphic to $\hat{\boldsymbol{\beta}}_{t+1} = \hat{\boldsymbol{\beta}}_t - \eta \mathbf{X}^\top (\mathbf{X} \hat{\boldsymbol{\beta}}_t - \mathbf{y})$ of Chapter 4, except that “inner products of the design matrix” are replaced by “inner products of parameter gradients.”

The astonishing fact about the NTK is its **wide-network limit**: with $\frac{1}{\sqrt{D}}$ scaling (as in (47)), as the width $D \rightarrow \infty$:

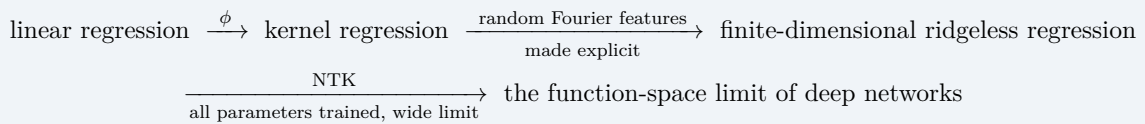
- (1) at initialization, Θ_0 converges by the law of large numbers to a **deterministic kernel** Θ (for two-layer networks it can be computed explicitly; see Exercise 8.1);
- (2) throughout **the entire training**, Θ_t stays put ($\Theta_t \rightarrow \Theta$, with no drift in θ_t) — parameters travel far, the kernel never moves;
- (3) hence (49) becomes a **linear ODE** $\frac{df_t(z)}{dt} = -\sum_i \Theta(z, z_i) r_{t,i}$, whose solution converges exponentially along the eigendirections of Θ , with limit

$$f_\infty(\tilde{z}) = \Theta(\tilde{z})^\top \Theta^{-1} \mathbf{y}, \quad (51)$$

where $\Theta = (\Theta(z_i, z_j))_{ij}$ is the Gram matrix of the NTK at the training points and $\Theta(\tilde{z})$ is the kernel vector of the new sample.

Equation (51) is **isomorphic to (43) symbol by symbol**: the function-space limit, under gradient descent, of an infinitely wide two-layer (indeed arbitrarily deep[25]) neural network with all parameters trained, is exactly **ridgeless kernel regression with the NTK as kernel**. The nonconvexity of parameter space is an “illusion of the coordinate system” — switch to function space, and deep-network training is Chapter 5’s minimum-norm interpolation story all over again.

The bridge from a straight line to deep networks



The four arrows are supported, respectively, by Section 8.3 (push-through), (48) (the law of large numbers), Section 5.5 (implicit regularization), and (51) (NTK constancy). **Linear regression is not a special case of deep learning; the wide limit of deep learning is the kernelization of linear regression** — which is why a two-hundred-year-old straight line remains the most honest starting point for understanding every model today.

Two pragmatic remarks. First, the NTK describes the *lazy training* regime: large parameter-initialization scale, small training displacement — the network is “too lazy” to change its internal features. When feature learning actually happens in real deep learning, the NTK fails, and theory is still under construction (Jacot’s own subsequent work distinguishing the lazy regime from the feature-learning regime is exactly about this). Second, ridgeless interpolation is **not only harmless — it can be better**: Belkin et al.’s double-descent observation shows that the test error of modern overparameterized models keeps decreasing past the interpolation threshold[26] — the minimum-norm solution of Section 5.3 is the cleanest linear incarnation of this

phenomenon, and work of Jingfeng Wu et al. further shows that optimally early-stopped GD statistically dominates optimally tuned ridge (see the preface and the bibliography)[28]. Put these two remarks next to (51), and you hold the most honest linear answer currently available to “why do overparameterized networks generalize?”

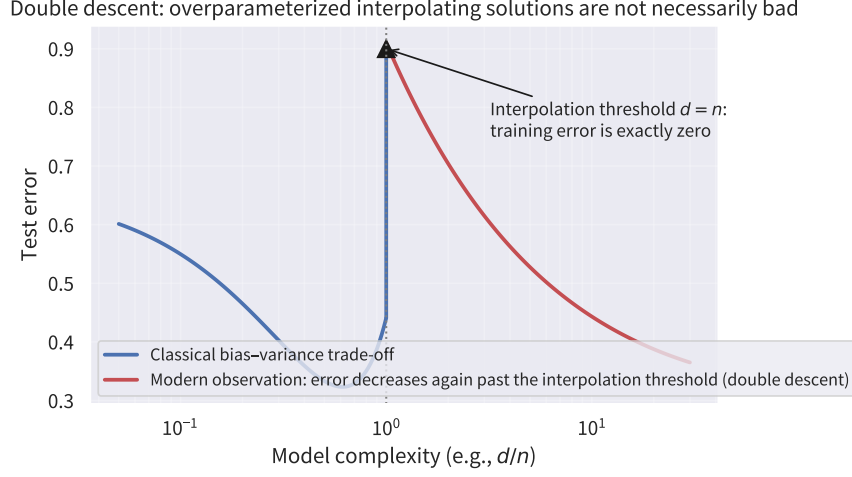


图 11: Double descent (schematic; Belkin et al., 2019[26]): classical textbooks say test error is optimal near $d \approx n$ (blue, U-shape); modern observation shows the error spikes at the interpolation threshold $d = n$ and then **keeps decreasing** (red) — interpolation itself is no sin; what matters is *which* interpolating solution implicit regularization selects. Ridgeless regression and early-stopped GD give the linear account of this selection.

Exercise 8.1. For the two-layer network (47) (smooth σ), let $(\omega_j, b_j) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ i.i.d. and $\mathbf{a}$ be initialized independently with $\mathbb{E}[a_j^2] = 1$. Verify: as $D \rightarrow \infty$, the NTK at initialization (50) converges pointwise to

$$\Theta(\mathbf{z}, \mathbf{z}') = \mathbb{E}[\sigma(\omega^\top \mathbf{z} + b) \sigma(\omega^\top \mathbf{z}' + b)] + (\mathbf{z}^\top \mathbf{z}' + 1) \mathbb{E}[\sigma'(\omega^\top \mathbf{z} + b) \sigma'(\omega^\top \mathbf{z}' + b)],$$

and identify which term comes from the gradients of the last-layer parameters and which from the first-layer parameters; then explain why the second term (absent from the random-feature kernel (48)) makes the NTK strictly stronger than the random-feature kernel. For $\sigma = \text{ReLU}$, use the half-space moment formulas of the Gaussian distribution to write both expectations explicitly in terms of the angle between $\mathbf{z}$ and $\mathbf{z}'$.

8.6 Conclusion of this chapter

Takeaways

- With $\mathbf{x}_i = \phi(\mathbf{z}_i)$, writing $\mathbf{K}_{ij} = k(\mathbf{z}_i, \mathbf{z}_j)$ and $\mathbf{k}(\tilde{\mathbf{z}})_i = k(\tilde{\mathbf{z}}, \mathbf{z}_i)$, we have

$$d > n : \tilde{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}' = \mathbf{k}(\tilde{\mathbf{z}})^\top \mathbf{K}^{-1} \mathbf{y} \xrightarrow{\lambda} n \gg d : \tilde{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}^{\text{ridge}} = \mathbf{k}(\tilde{\mathbf{z}})^\top (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y}.$$

Both closed-form solutions of linear regression are **estimators assembled purely from the kernel function**;

- The two key steps of kernelization: inner-product-ization (write every quantity as $\langle \phi, \phi \rangle$) and push-through $((\Phi^\top \Phi + \lambda \mathbf{I})^{-1} \Phi^\top = \Phi^\top (\mathbf{K} + \lambda \mathbf{I})^{-1})$ — the latter flips the inverse from feature space into sample space;
- The prediction $\hat{f}(\tilde{\mathbf{z}}) = \sum_i \alpha_i k(\tilde{\mathbf{z}}, \mathbf{z}_i)$ is **nonlinear** in the latent variables $\mathbf{z}$, even though it is linear in $\mathbf{x} = \phi(\mathbf{z})$ — “the prediction of linear regression is a natural kernel method under nonlinear latent-space mappings” is now proved;
- At $d = \infty$ (Gaussian kernel) kernelization goes from convenience to necessity; random features go the other way and approximate the kernel back into finite-dimensional linear algebra, welding the route back to parameterized networks.

Exercise 8.2. Use (45) to prove that the primal and dual forms of kernel ridge regression give **exactly the same** prediction function, and explain why no rank assumption is needed when $\lambda > 0$.

Exercise 8.3. Verify that the Gaussian kernel matrix $\mathbf{K}$ (with distinct samples) is positive definite and invertible, so that (43) is well defined; further verify that $\boldsymbol{\alpha}$ in (44) satisfies $\mathbf{K}\boldsymbol{\alpha} = \mathbf{y}$ (the kernel interpolation equation), and connect it with the interpolation verification (25) of Chapter 5.

9 Epilogue: So, Why Linear Regression?

Gather the eight threads.

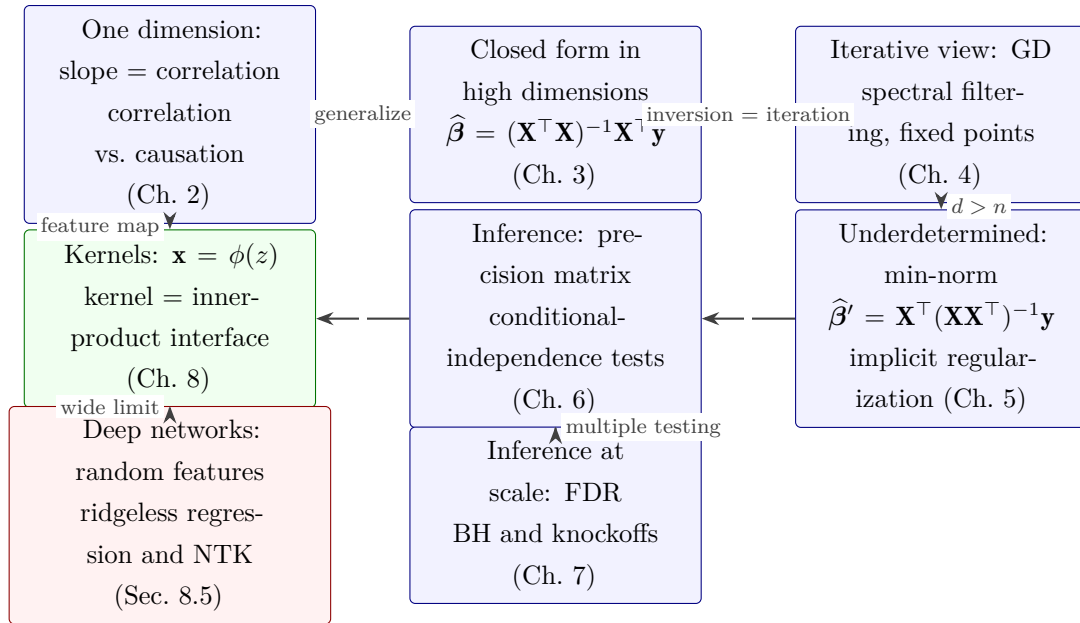


图 12: The roadmap of this book: starting from one straight line, proceeding along the two routes of closed form and iteration, converging in the underdetermined case to minimum-norm interpolation, upgrading through the precision matrix into per-coefficient inference, then through multiple testing into inference at scale (how much of a page of conclusions can be trusted), then kernelizing through the feature map, and finally bridging to the wide limit of deep networks.

The one-dimensional thread: least squares generalizes “the mean” into “the line”; the slope $\hat{b} = r \sigma_y / \sigma_x$ lets the correlation coefficient emerge naturally from geometric optimization; and the variance decomposition $\text{Var}(y) = b^2 \text{Var}(x) + \text{Var}(\varepsilon)$ shows that correlation is, in essence, a transfer of variance. The same straight line exposes the *symmetry* of the correlation measure (r) versus the *directionality* of the regression operation ($y \sim x$ and $x \sim y$ are two lines), and quantifies Galton’s “regression toward the mean” through the product $r^2 < 1$ of the two regression coefficients. And the causal direction — $x \rightarrow y$, $y \rightarrow x$, or a common cause — generates identical correlation structure and can only be identified outside the data[6, 7].

The high-dimensional thread: when $n \gg d$, least squares gives a unique, global, closed-form solution $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, the orthogonal projection of $\mathbf{y}$ onto the column space, and — under extremely weak error assumptions — the best linear unbiased estimator in the Gauss–Markov sense[3]. As n approaches d , the variance breaks first and the rank breaks second; ridge and lasso trade a budget of bias back for variance, letting the same framework extend into modern high-dimensional statistics[9, 10].

The optimization thread: the closed form $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ and gradient descent $\hat{\beta}_{t+1} = \hat{\beta}_t - \eta \mathbf{X}^\top (\mathbf{X} \hat{\beta}_t - \mathbf{y})$ are two paths to the same object — the fixed-point condition $\nabla L = \mathbf{0}$ is the normal equation; combining the two, the iterate has the exact expression $\hat{\beta}_t =$

$\hat{\beta} + (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t (\hat{\beta}_0 - \hat{\beta})$, where the convergence condition $0 < \eta < 2/\lambda_{\max}$ and the optimal rate $(\kappa - 1)/(\kappa + 1)$ promote “standardize your features” from folklore to half-theorem (it removes only the unit-driven part of ill-conditioning). Inverting in one step, and mortgaging the inverse with a t -term polynomial, are two faces of one geometry.

The underdetermined thread: when $d > n$, solutions are no longer unique but form the manifold $\hat{\beta}' + \mathcal{N}(\mathbf{X})$, on which every point is an interpolating solution and a GD fixed point; the manifold attracts transversally while individual points are only Lyapunov stable, and the landing point is decided by the null-space component of the initialization. $\hat{\beta}' = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{y}$ is the unique minimum-norm point on the manifold — exactly the pseudoinverse solution $\mathbf{X}^+ \mathbf{y}$; GD starting from zero converges to it implicitly, and ridge with $\lambda \rightarrow 0^+$ converges to it too. *The five faces (two closed forms, the pseudoinverse, the GD-from-zero limit, the ridge zero limit) are one and the same minimum-norm least-squares solution* — the linear prototype of “regularization without penalty” in the overparameterized era.

The inference thread: regression coefficients are elements of the precision matrix, $\beta_j = 0 \Leftrightarrow y \perp x_j \mid \mathbf{x}_{-j}$ — the one-dimensional “correlation vs. causation” graduates into the conditional-independence language of graphical models[15]; lasso, elastic net, graphical lasso, and CLIME deliver sparse estimation under $d \gg n$, and the debiased lasso upgrades point estimates into asymptotically normal inference — *estimation answers “which edges might exist,” inference answers “is this edge significant”*[17, 18, 19, 20, 21, 22].

The FDR thread: beyond the p -value of a single coefficient, the applied scientist faces a whole page of simultaneously tested conclusions. Bonferroni crushes the threshold to α/m and power collapses with m ; Benjamini–Hochberg trades a slanted line qk/m for “tolerate a few false ones when there are many discoveries,” controlling the expected false discovery proportion at q [31]; more satisfyingly, the BH threshold is exactly an adaptive sparse estimator of the z -scores, sharing its roots with the soft thresholding of Chapter 6 — *controlling the false discovery rate is not a patch on top of estimation; it is the dual of the sparsity assumption on the testing problem*[32]. Knockoffs take a more radical route: planting shadow variables inside the design matrix, they control the FDR of variable selection directly through exchangeability symmetry, without relying on any asymptotic normality of p -values[30]. *Estimation says which edges might exist, inference says which edges are significant, and FDR says how many of this page to trust.*

The kernel thread: when the observed features are nonlinear projections $\mathbf{x}_i = \phi(\mathbf{z}_i)$ of latent variables, the predictions of the two closed forms — $\tilde{\mathbf{x}}^\top \hat{\beta}' = \mathbf{k}(\tilde{\mathbf{z}})^\top \mathbf{K}^{-1} \mathbf{y}$ for $d > n$ and $\tilde{\mathbf{x}}^\top \hat{\beta}^{\text{ridge}} = \mathbf{k}(\tilde{\mathbf{z}})^\top (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y}$ for $n \gg d$ — are assembled solely from the kernel function: linear in the features $\Leftrightarrow$ nonlinear in latent space, and kernelization with push-through are two sides of the same coin[12, 13, 14].

The historical thread: from the appendix Legendre and Gauss wrote for the orbit of Ceres[1, 2], through Galton’s height scatterplots[4], to every regression output layer today — what has survived all of it is not a trick but a theorem about “how data meets model.”

The final answer

Because linear regression is the **only** model that simultaneously does all four of the following:

- (i) **Exactly solvable:** a convex quadratic problem, closed-form solution, milliseconds, no hyperparameters, no local minima;
- (ii) **Interpretable:** every coefficient has something to say; R^2 , leverage, and degrees of freedom each have a geometric meaning;
- (iii) **Auditable:** every theorem carries an explicit assumption list, and when the assumptions fail there are named diagnostics and repairs;
- (iv) **Generalizable:** change the loss and get robust regression; add a penalty and get ridge/lasso; add features and get nonlinearity; iterate with weights and get GLMs; stack training and get neural networks.

Fit the straight line first, audit the assumptions, disclose the weaknesses honestly — and only then, and only after that, do fancier models get their turn.

Exercise 9.1. Suppose $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$ (orthogonal design). Find the λ^* minimizing $\text{MSE}(\hat{\beta}_j^{\text{ridge}})$, and show that $\lambda^* > 0$ if and only if the coefficient is a “weak signal”; use this to state precisely what “shrinkage never hurts a strong signal and always helps a weak one” means.

A Solutions to Exercises

A.1 Chapter 3: High-Dimensional Linear Regression

Exercise 3.1 (completing the Gauss–Markov proof)

Step 1 (unbiasedness $\Rightarrow$ a linear constraint). $\mathbb{E}[\tilde{\beta}] = \mathbf{A}\mathbf{X}\beta = \beta$ for all β , so $\mathbf{A}\mathbf{X} = \mathbf{I}$. Take $\mathbf{B} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ (the OLS coefficient matrix, $\mathbf{B}\mathbf{X} = \mathbf{I}$) and let $\mathbf{D} := \mathbf{A} - \mathbf{B}$; then

$$\mathbf{D}\mathbf{X} = \mathbf{A}\mathbf{X} - \mathbf{B}\mathbf{X} = \mathbf{I} - \mathbf{I} = \mathbf{0}.$$

Conversely, for any $\mathbf{D}$ with $\mathbf{D}\mathbf{X} = \mathbf{0}$, $(\mathbf{B} + \mathbf{D})\mathbf{y}$ is unbiased — so “unbiased linear estimators” and “OLS plus a null-space perturbation” are the same family of objects.

Step 2 (variance comparison). From $\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}$,

$$\text{Var}(\tilde{\beta}) = \sigma^2 (\mathbf{B} + \mathbf{D})(\mathbf{B} + \mathbf{D})^\top = \sigma^2 \mathbf{B}\mathbf{B}^\top + \sigma^2 (\mathbf{B}\mathbf{D}^\top + \mathbf{D}\mathbf{B}^\top) + \sigma^2 \mathbf{D}\mathbf{D}^\top.$$

The cross terms vanish: $\mathbf{D}\mathbf{B}^\top = \mathbf{D}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{0}$, and transposing gives $\mathbf{B}\mathbf{D}^\top = \mathbf{0}$. Hence

$$\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}) = \sigma^2 \mathbf{D}\mathbf{D}^\top \succeq \mathbf{0},$$

since for any $\mathbf{v}$, $\mathbf{v}^\top \mathbf{D}\mathbf{D}^\top \mathbf{v} = \|\mathbf{D}^\top \mathbf{v}\|^2 \geq 0$. Equality holds for all $\mathbf{v}$ if and only if $\mathbf{D} = \mathbf{0}$, i.e., $\tilde{\beta} = \hat{\beta}$. $\square$

Exercise 3.2 (the bias of ridge regression)

First-order condition: $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}) \hat{\beta}^{\text{ridge}} = \mathbf{X}^\top \mathbf{y}$, i.e., $\hat{\beta}^{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ (note $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ is positive definite for any $\lambda > 0$; no full-rank assumption needed). Taking expectations and using $\mathbb{E}[\mathbf{y}] = \mathbf{X}\beta$:

$$\begin{aligned} \mathbb{E}[\hat{\beta}^{\text{ridge}}] &= (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{X} \beta \\ &= \beta - \lambda (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \beta, \end{aligned}$$

where the second step splits $\mathbf{X}^\top \mathbf{X}$ into $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}) - \lambda \mathbf{I}$ so the inverse cancels against itself. Hence the bias $= -\lambda (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \beta$: the direction is “pulled toward the origin,” and the magnitude is discounted direction-by-direction by $1/(\text{eigenvalue} + \lambda)$ — consistent with the SVD analysis in the main text. $\square$

A.2 Chapter 4: Gradient Descent, an Optimization View

Exercise 4.1 (convergence bound under the optimal step size)

Let $\mathbf{A} = \mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}$. From (21) and $\hat{\beta}_0 = \mathbf{0}$:

$$\hat{\beta}_t - \hat{\beta} = \mathbf{A}^t (\mathbf{0} - \hat{\beta}) = -\mathbf{A}^t \hat{\beta}.$$

$\mathbf{A}$ is symmetric, so $\|\mathbf{A}^t\|_2 = \|\mathbf{A}\|_2^t = (\max_j |1 - \eta \lambda_j|)^t$. Substituting $\eta = \eta^*$ (Corollary 4.2), $\max_j |1 - \eta^* \lambda_j| = \frac{\kappa - 1}{\kappa + 1} =: \rho^*$, and therefore

$$\|\hat{\beta}_t - \hat{\beta}\| \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^t \|\hat{\beta}\|.$$

To reach $\|\hat{\beta}_t - \hat{\beta}\| \leq \varepsilon$ we need $(\rho^*)^t \leq \varepsilon / \|\hat{\beta}\|$, i.e.,

$$t \geq \frac{\log(\|\hat{\beta}\|/\varepsilon)}{\log(1/\rho^*)}.$$

When $\kappa \gg 1$, $\log \frac{1}{\rho^*} = \log \frac{\kappa+1}{\kappa-1} \approx \frac{2}{\kappa}$, so $t = O(\kappa \log \frac{\|\hat{\beta}\|}{\varepsilon})$: the number of steps grows **linearly** with the condition number. $\square$

Exercise 4.2 (the Neumann identity)

Let $\mathbf{A} = \mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}$; then $\eta \mathbf{X}^\top \mathbf{X} = \mathbf{I} - \mathbf{A}$, and by the normal equation $\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{y}$. Using the geometric-series identity $\mathbf{I} - \mathbf{A}^t = (\mathbf{I} - \mathbf{A}) \sum_{k=0}^{t-1} \mathbf{A}^k$ (expand and telescope; no convergence needed):

$$[\mathbf{I} - \mathbf{A}^t] \hat{\beta} = \eta \mathbf{X}^\top \mathbf{X} \sum_{k=0}^{t-1} \mathbf{A}^k \hat{\beta} = \eta \sum_{k=0}^{t-1} \mathbf{A}^k \mathbf{X}^\top \mathbf{y},$$

where the last step merges the adjacent $\mathbf{X}^\top \mathbf{X}$ and $\hat{\beta}$ into $\mathbf{X}^\top \mathbf{y}$. The left-hand side is exactly the left-multiplied form $\hat{\beta}_t = [\mathbf{I} - \mathbf{A}^t] \hat{\beta}$ of (22) — the t -step GD solution always equals “applying the t -th partial-sum approximation of the inverse” to $\hat{\beta}$. Note this identity is exact for every finite t , regardless of the size of η ; convergence ($\mathbf{A}^t \rightarrow \mathbf{0}$) is what requires $0 < \eta < 2/\lambda_{\max}$. $\square$

A.3 Chapter 5: The Underdetermined Case and Implicit Regularization

Exercise 5.1 (the primal–dual identity of ridge, and the zero limit)

The identity. Multiply both sides on the left by $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})$:

$$(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}) \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I})^{-1} = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}) (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I})^{-1} = \mathbf{X}^\top,$$

(the first step uses $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}) \mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{X}^\top + \lambda \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I})$, plain distributivity of matrix multiplication.) which equals the left-hand side’s $\mathbf{X}^\top$; hence the identity holds (both matrices are well defined, since $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ and $\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}$ are positive definite for $\lambda > 0$).

The zero limit. Substitute the SVD $\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$ (singular values s_j); the j -th spectral component of the identity acting on $\mathbf{y}$ is

$$\frac{s_j}{s_j^2 + \lambda} (\mathbf{U}^\top \mathbf{y})_j \xrightarrow{\lambda \rightarrow 0^+} \begin{cases} \frac{1}{s_j} (\mathbf{U}^\top \mathbf{y})_j, & s_j > 0, \\ 0, & s_j = 0, \end{cases}$$

and the right-hand side is exactly $\mathbf{V} \mathbf{\Sigma}^+ \mathbf{U}^\top \mathbf{y} = \mathbf{X}^+ \mathbf{y}$. No full-rank assumption was used anywhere — this is a direct proof that “ridge’s zero limit = the pseudoinverse solution,” and also a special case of the push-through identity (45). $\square$

Exercise 5.2 (the GD limit from an arbitrary initialization)

The general solution of iteration (18) (direct induction for a linear iteration):

$$\hat{\beta}_t = A^t \hat{\beta}_0 + \eta \sum_{k=0}^{t-1} A^k X^\top y, \quad A = I - \eta X^\top X.$$

First term. A is the identity on $\mathcal{N}(X)$ and has spectral radius < 1 on $\text{row}(X)$ (for $0 < \eta < 2/\lambda_{\max}(XX^\top)$), so $A^t \rightarrow P_{\mathcal{N}(X)} = I - X^+ X$ (the orthogonal projection onto the null space).

Second term. As in the proof of Theorem 5.2: $\eta \sum_{k < t} A^k X^\top \rightarrow X^\top (XX^\top)^{-1}$, i.e., $\hat{\beta}'$.

Combining. $\lim_t \hat{\beta}_t = \hat{\beta}' + P_{\mathcal{N}(X)} \hat{\beta}_0$. The null-space component of the initialization is carried along the whole way (“null-space neutrality” of Section 5.4), while the row-space component converges to the nearest point on the manifold. When $\hat{\beta}_0 = 0$, $P_{\mathcal{N}(X)} 0 = 0$, and we recover Theorem 5.2 — the two statements agree completely. $\square$

Exercise 5.3 (the nonzero spectra of $X^\top X$ and $XX^\top$ coincide)

Let $X^\top X v = \lambda v$ with $\lambda \neq 0$, $v \neq 0$. First show $w := Xv \neq 0$: if $Xv = 0$ then $\lambda v = X^\top Xv = 0$, contradicting $\lambda \neq 0$, $v \neq 0$. Then

$$(XX^\top) w = X(X^\top Xv) = \lambda Xv = \lambda w,$$

so w is an eigenvector of $XX^\top$ with eigenvalue λ . The reverse direction swaps X and $X^\top$. The map $v \mapsto Xv$ is one-to-one between the eigenspaces (eigenvectors of distinct eigenvalues never mix), so **the nonzero eigenvalues and their multiplicities are exactly the same**; the counts of zero eigenvalues are respectively $d - r$ and $n - r$ ($r = \text{rank}(X)$) and may differ.

Since GD’s decay factor $1 - \eta\lambda_j$ acts only on the nonzero spectrum, the convergence-rate formulas of the two regimes depend only on the r eigenvalues that $X^\top X$ and $XX^\top$ share — the algebraic reason the “ d formulas” and the “ n formulas” are interchangeable: the effective dimension is r in both cases. $\square$

A.4 Chapter 6: From Estimation to Inference**Exercise 6.1 (the block-inversion trilogy)**

Write $\Sigma = \begin{pmatrix} s & c^\top \\ c & S \end{pmatrix}$ ($s = \Sigma_{yy}$, $c = \Sigma_{Xy}$, $S = \Sigma_{XX}$, with the relevant inverses assumed to exist).

(i) Conditional distribution. Apply the linear transformation $y \mapsto y - c^\top S^{-1}x$ to the joint Gaussian to eliminate the cross term; this immediately gives

$$y \mid X \sim \mathcal{N}(c^\top S^{-1}x, s - c^\top S^{-1}c).$$

(ii) **Precision blocks.** Expand $\Theta\Sigma = \mathbf{I}$ blockwise into four equations, or use the Schur-complement inversion formula:

$$\Theta_{yy} = \frac{1}{s - \mathbf{c}^\top \mathbf{S}^{-1} \mathbf{c}}, \quad \Theta_{Xy} = -\mathbf{S}^{-1} \mathbf{c} \Theta_{yy}.$$

Comparing with (i) gives (30): $\beta = \mathbf{S}^{-1} \mathbf{c} = -\Theta_{Xy}/\Theta_{yy}$, $\text{Var}(\varepsilon) = s - \mathbf{c}^\top \mathbf{S}^{-1} \mathbf{c} = 1/\Theta_{yy}$.

(iii) **Partial correlation.** Apply the same block decomposition to Θ itself (fix indices i, j , treat the rest as one block); the correlation of the two-dimensional Schur complement of i, j is the partial correlation: $\rho_{ij \cdot V \setminus \{i, j\}} = -\Theta_{ij}/\sqrt{\Theta_{ii}\Theta_{jj}}$. In particular $\Theta_{ij} = 0$ if and only if this partial correlation vanishes, and by the argument of (ii) repeated for variable i instead of y , this is equivalent to $i \perp j \mid \text{the rest}$. $\square$

Exercise 6.2 (error decomposition of the debiased estimator)

From the model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, the residual is $\mathbf{y} - \mathbf{X}\hat{\beta}_{\text{lasso}} = \mathbf{X}(\beta - \hat{\beta}_{\text{lasso}}) + \varepsilon$. Substitute into (34) and use $\mathbf{S} = \mathbf{X}^\top \mathbf{X}/n$:

$$\begin{aligned} \tilde{\beta} &= \hat{\beta}_{\text{lasso}} + \frac{1}{n} \hat{\Theta} \mathbf{X}^\top \varepsilon + \hat{\Theta} \mathbf{S} (\beta - \hat{\beta}_{\text{lasso}}) \\ &= \beta + \underbrace{\frac{1}{n} \hat{\Theta} \mathbf{X}^\top \varepsilon}_{\text{noise (variance)}} + \underbrace{(\hat{\Theta} \mathbf{S} - \mathbf{I}) (\beta - \hat{\beta}_{\text{lasso}})}_{\text{remainder (bias)}}, \end{aligned}$$

where the second step uses the identity split $\hat{\beta}_{\text{lasso}} = \beta - (\beta - \hat{\beta}_{\text{lasso}})$.

Why the first term is “variance.” It is a mean-zero independent sum $\frac{1}{n} \sum_i \hat{\Theta} \mathbf{x}_i \varepsilon_i$; conditional on $\mathbf{X}$ (hence on $\hat{\Theta}$), each term has variance $\frac{\sigma^2}{n} \cdot \frac{1}{n} \hat{\Theta} \mathbf{X}^\top \mathbf{X} \hat{\Theta}^\top \approx \frac{\sigma^2}{n} \Theta \Sigma \Theta$ (using $\hat{\Theta} \mathbf{S} \approx \mathbf{I}$), and the central limit theorem gives the asymptotic normality (35). It *derives purely from noise*, regardless of how far $\hat{\beta}_{\text{lasso}}$ is from β .

Why the second term is “bias.” Written componentwise: the absolute value of the j -th component is at most $\|\hat{\Theta} \mathbf{S} - \mathbf{I}\|_{\max} \cdot \|\beta - \hat{\beta}_{\text{lasso}}\|_1$. The nodewise-lasso or CLIME construction guarantees $\|\hat{\Theta} \mathbf{S}\|_{\max} = O_p(\lambda_n)$ with $\lambda_n \asymp \sqrt{\log d/n}$; the lasso ℓ_1 error under s -sparsity satisfies $\|\beta - \hat{\beta}_{\text{lasso}}\|_1 = O_p(s\lambda_n)$. The product is $O_p(s\lambda_n^2) = O_p(s \log d/n) = o_p(n^{-1/2})$ (when $s \log d = o(\sqrt{n})$). Hence the bias term is an order of magnitude smaller than the noise term, and asymptotic normality follows — the essence of debiasing is that $\hat{\Theta}$ “projects out” the shrinkage bias. $\square$

Exercise 6.3 (points for discussion)

The optimality condition of graphical lasso, $\mathbf{S} - \Theta^{-1} + \lambda \Gamma = \mathbf{0}$, is *nonlinear* in Θ (it contains Θ^{-1}), and the whole iteration must stay inside the cone $\Theta \succ \mathbf{0}$; when $d \gg n$ and $\mathbf{S}$ is highly ill-conditioned, floating-point evaluation of Θ^{-1} **amplifies the ill-conditioning of $\mathbf{S}$ quadratically**, worsening the condition number of the optimization problem. CLIME rewrites “ Θ is the inverse of $\mathbf{S}$ ” as the linear constraint $\mathbf{S}\Theta - \mathbf{I} \approx \mathbf{0}$: the ill-conditioning enters only through one matrix multiplication, and geometrically the problem is a linear program over a polytope — better conditioned, no PD constraint, no

log det differentiation. Practical benefits of CLIME's per-column LP structure: (i) the d columns are fully independent, embarrassingly parallel; (ii) each column is a standard LP callable from any mature solver, with no need to implement glasso's block coordinate descent; (iii) it allows scenarios where Σ itself is not sparse and only Θ is (likelihood-based methods have a more tortuous efficiency theory there). The cost: CLIME discards the statistical efficiency of the likelihood and is often slightly more conservative than glasso in finite samples. $\square$

A.5 Chapter 7: Inference at Scale and FDR Control

Exercise 7.1 (BH by hand)

The threshold is $qk/m = 0.02k$: $k = 1$: 0.02 ($0.001 \leq 0.02$, pass); $k = 2$: 0.04 ($0.02 \leq 0.04$, pass); $k = 3$: 0.06 ($0.04 \leq 0.06$, pass); $k = 4$: 0.08 ($0.09 > 0.08$, fail); $k = 5$: 0.10 ($0.3 > 0.10$, fail). Hence $\hat{k} = 3$, and BH rejects $\{1, 2, 3\}$.

Three-way comparison: the Bonferroni threshold $q/m = 0.02$ rejects only $\{1\}$ — it misses two discoveries that could have been made; testing one by one (at level $q = 0.1$) rejects $\{1, 2, 3, 4\}$ — the fourth hypothesis, with $p = 0.09$, is rejected with no protection, even though it is very likely a null (with $m = 5$, the probability of seeing $p \leq 0.1$ under all-null is about 40%). BH sits exactly in between: two more discoveries than Bonferroni, one less risk than the naked test. Note that $0.09 < q$ yet it is *not* rejected by BH — **BH is not “per-hypothesis testing with α replaced by qk/m ”; the slanted line has already tightened beyond the 4-th p -value.** $\square$

Exercise 7.2 (FDR vs. FWER; the inclusion between the two procedures)

(a) Examine the ratio sample-path by sample-path: if $V = 0$, then $V/\max(R, 1) = 0$; if $V \geq 1$, then $V/\max(R, 1) \leq 1 = \mathbf{1}\{V \geq 1\}$. Taking expectations,

$$\text{FDR} = \mathbb{E}\left[\frac{V}{\max(R, 1)}\right] \leq \mathbb{E}[\mathbf{1}\{V \geq 1\}] = \mathbb{P}(V \geq 1) = \text{FWER}.$$

So any procedure with FWER controlled at α (Bonferroni, Holm, ...) automatically satisfies $\text{FDR} \leq \alpha$ — usually far too conservatively.

(b) Bonferroni (at level q) rejects $\{j : p_j \leq q/m\}$. Suppose some $p_j = p_{(i)}$ lies in this set; then $p_{(i)} \leq q/m \leq qi/m$ (since $i \geq 1$), i.e., i satisfies the crossing condition (37), so $i \leq \hat{k}$ and the i -th hypothesis is also rejected by BH. Hence the Bonferroni rejection set $\subseteq$ the BH rejection set, a strict subset whenever $\hat{k} \geq 2$.

(c) Bonferroni's per-test threshold: two-sided $q/m = 5 \times 10^{-7}$, corresponding to $|z| > z_{1-2.5 \times 10^{-7}} \approx 5.17$. The power of a single signal ($z_j \sim \mathcal{N}(5, 1)$) is

$$\mathbb{P}(|z_j| > 5.17) = \Phi(5 - 5.17) + \Phi(-5 - 5.17) \approx \Phi(-0.17) \approx 0.43.$$

About $200 \times 0.43 \approx 86$ expected rejections — power crushed to four tenths, collapsing further as m grows at fixed signal strength.

BH: the signals' p -values concentrate near $2\Phi(-5) \approx 5.7 \times 10^{-7}$, three orders of magnitude below the threshold. Take $\hat{k} \approx 210$ (200 signals + about 10 null p -values mixed in); the threshold $q\hat{k}/m = 0.05 \times 210/10^5 \approx 1.05 \times 10^{-4}$ is $\hat{k} \approx 210$ times the Bonferroni threshold; all signals pass and power $\rightarrow 1$. Verify the false discovery proportion: of about 210 rejections, about 10 are false, $10/210 \approx 4.8\% \leq q = 5\%$ — consistent with Theorem 7.1.

□

Exercise 7.3 (induction proof of $\text{FDR} \leq q$ under the all-null independent case)

Let $P_m(q) = \mathbb{P}(\exists k \leq m : p_{(k)} \leq qk/m)$ with p_j iid $U(0, 1)$. **Base case** $m = 1$: $P_1(q) = \mathbb{P}(p_1 \leq q) = q$.

Inductive step: assume the claim for $m - 1$. Condition on the maximum $U := p_{(m)}$, whose density is mu^{m-1} , splitting into two branches.

Branch one: $U \leq q$. Then $k = m$ automatically satisfies the crossing condition ($U \leq q = qm/m$), the event holds with probability one, contributing

$$\int_0^q 1 \cdot mu^{m-1} du = q^m.$$

Branch two: $U = u > q$. Key fact: conditional on $U = u$, the remaining $m - 1$ variables are iid $U(0, u)$, i.e., $p_{(k)} = u q_{(k)}$ ($k < m$), where $q_{(k)}$ are the order statistics of $m - 1$ uniforms. Hence

$$p_{(k)} \leq \frac{qk}{m} \iff q_{(k)} \leq \frac{qk}{mu} = \frac{q'k}{m-1}, \quad q' := \frac{q(m-1)}{mu}.$$

By the inductive hypothesis (applied to $m - 1$ iid uniforms at level q' ; the conclusion is trivial when $q' > 1$), the conditional probability in this branch is $\leq q' = q(m-1)/(mu)$, contributing

$$\int_q^1 \frac{q(m-1)}{mu} mu^{m-1} du = q(m-1) \int_q^1 u^{m-2} du = q(1 - q^{m-1}).$$

Combining: $P_m(q) \leq q^m + q(1 - q^{m-1}) = q$. Induction complete. □

(Note: the general case $m_0 < m$ uses conditioning on the non-null p -values plus the technique of positive regression dependence (PRDS) to reduce to the all-null case; the original Benjamini–Hochberg paper and Benjamini–Yekutieli (2001) give the full argument.)

Exercise 7.4 (the exchangeability argument)

(a) Let the augmented design be $\mathbf{A} = [\mathbf{X} \ \tilde{\mathbf{X}}] \in \mathbb{R}^{n \times 2d}$. The defining construction property of knockoffs is exactly: for any j , swapping column j with column $d + j$ of $\mathbf{A}$ leaves the joint distribution of $\mathbf{A}$ unchanged. Under H_{0j} , $\mathbf{y}$ does not depend on $\mathbf{x}_j$ ($\mathbf{y} \perp \mathbf{x}_j \mid \mathbf{x}_{-j}$), so the swap does not change the joint distribution of $(\mathbf{A}, \mathbf{y})$, and hence does not change the distribution of any statistic computed from them. The idealized importance statistic satisfies **swap equivariance**: under the swap, $(Z_j, \tilde{Z}_j) \rightarrow (\tilde{Z}_j, Z_j)$ with all other components fixed, and therefore

$$(W_1, \dots, W_j, \dots, W_m) = (W_1, \dots, -W_j, \dots, W_m) \text{ in distribution.}$$

Marginalizing out the other coordinates gives $W_j \stackrel{d}{=} -W_j$. This is (38). *An honest note:* in actual statistics, the other W_i also shift slightly because the shadow $\tilde{\mathbf{x}}_j$ participates in the fit; the rigorous argument needs the pair “exchangeability of the augmented design + equivariance of the statistic under swaps” (the sign-flip property of Barber–Candès 2015), and the conclusion is unchanged: under the nulls, the signs of the entire vector W can be flipped arbitrarily.

(b) The **discoveries** above a threshold t are $\{j : W_j \geq t\}$, with count $R(t) = \#\{j : W_j \geq t\}$. The number of false discoveries (those with H_{0j} true) $V(t)$ is unobservable, but by (a), each false discovery has W_j positive with probability $1/2$ and negative with probability $1/2$, independently of $|W_j|$ (by symmetry). Hence

$$\mathbb{E}[\#\{j \text{ null} : W_j \leq -t\}] = \mathbb{E}[\#\{j \text{ null} : W_j \geq t\}] = \mathbb{E}V(t),$$

i.e., the **observable count** on the left (contaminated by a few negative W ’s of signal variables, which only makes it larger and more conservative) estimates $\mathbb{E}V(t)$. The knock-off+ threshold (39) requires $(1 + \#\{W_j \leq -t\})/\#\{W_j \geq t\} \leq q$: the denominator is $R(t)$, the numerator is $\widehat{V}(t) + 1$ (the $+1$ counters the overdispersion of the numerator). Thus every candidate threshold satisfies $\widehat{V}(t)/R(t) \leq q$, and taking expectations over the random ordering (the technical part of the knockoff theorem is dressing this “ratio estimator” in a martingale argument) yields $\text{FDR} \leq q$. The intuition in one sentence: **the shadow variables are symmetric mirrors of the real ones, and the count of negative W ’s is the reflection of the fakes mixed in.** $\square$

A.6 Chapter 8: Nonlinear Latent Spaces

Exercise 8.2 (primal form = dual form of kernel ridge regression)

Primal-form prediction: $\hat{f}(\tilde{\mathbf{z}}) = \phi(\tilde{\mathbf{z}})^\top (\Phi^\top \Phi + \lambda \mathbf{I}_d)^{-1} \Phi^\top \mathbf{y}$. Apply the push-through identity (45) to $(\Phi^\top \Phi + \lambda \mathbf{I}_d)^{-1} \Phi^\top$ (with $\mathbf{X}$ there replaced by Φ):

$$\hat{f}(\tilde{\mathbf{z}}) = \phi(\tilde{\mathbf{z}})^\top \Phi^\top (\mathbf{K} + \lambda \mathbf{I}_n)^{-1} \mathbf{y} = \mathbf{k}(\tilde{\mathbf{z}})^\top (\mathbf{K} + \lambda \mathbf{I}_n)^{-1} \mathbf{y},$$

which is exactly the dual form (46) — the two give **exactly the same** prediction function pointwise.

Rank assumptions. When $\lambda > 0$, all eigenvalues of $\Phi^\top \Phi + \lambda \mathbf{I}_d$ and $\mathbf{K} + \lambda \mathbf{I}_n$ are $\geq \lambda > 0$, regardless of the rank of Φ — this is precisely the mechanism by which ridge regularization “incidentally” completes the kernelization: both inverses always exist. $\square$

Exercise 8.3 (positive-definiteness of the Gaussian kernel and the interpolation equation)

Strict positive-definiteness. The Fourier transform of the Gaussian kernel $k(\mathbf{z}, \mathbf{z}') = \exp(-\|\mathbf{z} - \mathbf{z}'\|^2 / 2\sigma^2)$ is a positive function (supported on the whole frequency space, no spectral gap). Suppose $\sum_i a_i k(\mathbf{z}, \mathbf{z}_i) \equiv 0$: the left side is a linear combination of n distinct Gaussian functions, hence an analytic function; a Fourier-support argument (or the

identity theorem for analytic functions) forces all $a_i = 0$. Hence $\{k(\cdot, \mathbf{z}_i)\}_{i=1}^n$ are linearly independent in the RKHS, their Gram matrix $\mathbf{K}$ is **positive definite and invertible**, and (43) is well defined.

The interpolation equation. By definition $\boldsymbol{\alpha} = \mathbf{K}^{-1}\mathbf{y}$, so $\mathbf{K}\boldsymbol{\alpha} = \mathbf{y}$. Verify interpolation: at a training point $\mathbf{z}_j$, $\mathbf{k}(\mathbf{z}_j)$ is the j -th row of $\mathbf{K}$, so

$$\hat{f}(\mathbf{z}_j) = \mathbf{k}(\mathbf{z}_j)^\top \mathbf{K}^{-1}\mathbf{y} = (\mathbf{K}e_j)^\top \mathbf{K}^{-1}\mathbf{y} = y_j.$$

Correspondence: this is the kernel-language translation of $\mathbf{X}\hat{\boldsymbol{\beta}}' = \mathbf{y}$ from Chapter 5 — feature-space interpolation $\phi(\mathbf{z}_j)^\top \hat{\boldsymbol{\beta}}' = y_j$, after inner-product-ization, becomes $\mathbf{K}\boldsymbol{\alpha} = \mathbf{y}$; the “coefficients $\hat{\boldsymbol{\beta}}'$ ” and the “kernel weights $\boldsymbol{\alpha}$ ” are linked by $\hat{\boldsymbol{\beta}}' = \boldsymbol{\Phi}^\top \boldsymbol{\alpha}$ (the representer form). $\square$

Exercise 8.1 (explicit form of the two-layer NTK)

Write $u = \boldsymbol{\omega}^\top \mathbf{z} + b$, $u' = \boldsymbol{\omega}^\top \mathbf{z}' + b$ (with $\boldsymbol{\omega}, b$ drawn from the initialization distribution of Section 8.5). For $f(\mathbf{z}) = \frac{1}{\sqrt{D}} \sum_{j=1}^D a_j \sigma(\boldsymbol{\omega}_j^\top \mathbf{z} + b_j)$ from (47), differentiate with respect to the two kinds of parameters:

$$\frac{\partial f}{\partial a_j} = \frac{1}{\sqrt{D}} \sigma(\boldsymbol{\omega}_j^\top \mathbf{z} + b_j), \quad \nabla_{(\boldsymbol{\omega}_j, b_j)} f = \frac{a_j}{\sqrt{D}} \sigma'(\boldsymbol{\omega}_j^\top \mathbf{z} + b_j) (\mathbf{z}^\top, 1)^\top.$$

The parameter inner product (under the $1/D$ scaling convention) splits into the sum of the two layers' contributions:

$$\Theta^{(D)}(\mathbf{z}, \mathbf{z}') = \frac{1}{D} \sum_{j=1}^D \left[\sigma(u_j) \sigma(u'_j) + a_j^2 \sigma'(u_j) \sigma'(u'_j) (\mathbf{z}^\top \mathbf{z}' + 1) \right].$$

As $D \rightarrow \infty$, the initialized parameters $a_j, \boldsymbol{\omega}_j, b_j$ are i.i.d., and the law of large numbers gives pointwise convergence to the expectation:

$$\Theta(\mathbf{z}, \mathbf{z}') = \mathbb{E}[\sigma(u) \sigma(u')] + \mathbb{E}[a^2] \cdot \mathbb{E}[\sigma'(u) \sigma'(u')] \cdot (\mathbf{z}^\top \mathbf{z}' + 1).$$

Under the standard initialization $\mathbb{E}[a^2] = 1$ this is

$$\Theta(\mathbf{z}, \mathbf{z}') = \underbrace{\mathbb{E}[\sigma(u) \sigma(u')]}_{\text{output-layer contribution (= RF kernel)}} + (\mathbf{z}^\top \mathbf{z}' + 1) \mathbb{E}[\sigma'(u) \sigma'(u')].$$

Two remarks: if the **biases are fixed** and not trained, the second term degenerates to $\mathbf{z}^\top \mathbf{z}' \mathbb{E}[\sigma' \sigma']$ (the form stated in the exercise uses this convention); if the biases are trained together with the other first-layer parameters, the inner product picks up an additional 1×1 , i.e., $\mathbf{z}^\top \mathbf{z}' + 1$ — this solution adopts the latter.

Comparison with the RF kernel. The RF kernel $k_{\text{RF}}(\mathbf{z}, \mathbf{z}') = \mathbb{E}[\sigma(u) \sigma(u')]$ contains only the output-layer contribution; the NTK has an extra derivative term, which at $\mathbf{z} = \mathbf{z}'$ satisfies $\mathbb{E}[\sigma'(u)^2] > 0$ and is strictly positive — hence $\Theta \succeq k_{\text{RF}}$ (as kernel matrices), strictly stronger than the RF kernel. $\square$

Explicit half-space formulas for ReLU. Take (ω, b) with independent standard normal components (equivalently, $(\omega_1, \dots, \omega_p, b)$ isotropic), and write $\varphi = \arccos(\mathbf{z}^\top \mathbf{z}' / (\|\mathbf{z}\| \|\mathbf{z}'\|))$ (ReLU networks are insensitive to constant shifts, so set $\|\mathbf{z}\| = \|\mathbf{z}'\| = 1$); then

$$\mathbb{E}[\text{ReLU}(u)\text{ReLU}(u')] = \frac{1}{2\pi} (\sin \varphi + (\pi - \varphi) \cos \varphi),$$

$$\mathbb{E}[\sigma'(u)\sigma'(u')] = \mathbb{E}[\mathbf{1}\{u > 0\} \mathbf{1}\{u' > 0\}] = \frac{\pi - \varphi}{2\pi}.$$

The latter is the angle measure of the intersection of two half-spaces, the former the classical half-space moment formula of Cho & Saul (2009). Substituting back gives the explicit NTK of a two-layer ReLU network. $\square$

A.7 Chapter 9: Epilogue

Exercise 9.1 (optimal shrinkage under an orthogonal design)

When $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$, the OLS coordinates are independent, $\hat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2/n)$, and the ridge solution shrinks coordinatewise:

$$\hat{\beta}_j^{\text{ridge}} = \frac{n}{n + \lambda} \hat{\beta}_j \implies \text{MSE}(\lambda) = \underbrace{\left(\frac{\lambda}{n + \lambda}\right)^2 \beta_j^2}_{\text{bias}^2} + \underbrace{\frac{\sigma^2 n}{(n + \lambda)^2}}_{\text{variance}}.$$

Differentiate with respect to λ and set to zero:

$$\frac{d}{d\lambda} \frac{\lambda^2 \beta_j^2 + \sigma^2 n}{(n + \lambda)^2} = 0 \iff \lambda \beta_j^2 (n + \lambda) = \lambda^2 \beta_j^2 + \sigma^2 n \iff \boxed{\lambda^* = \frac{\sigma^2}{\beta_j^2}}.$$

(When $\beta_j = 0$ the bias term is identically zero and the variance increases with λ , so the optimum is $\lambda^* = \infty$ — shrink the coefficient all the way to 0. Hence $\lambda^* > 0$ if and only if $\beta_j \neq 0$.)

The precise meaning of “weak signal.” Measure the signal in noise units $\sigma/\sqrt{n}$: $|\beta_j| \leq \sigma/\sqrt{n}$ (signal-to-noise ratio at most 1) if and only if $\lambda^* = \sigma^2/\beta_j^2 \geq n$, i.e., the shrinkage factor $n/(n + \lambda^*) \leq \frac{1}{2}$ — *the definition of a weak signal is “the optimal shrinkage cuts at least half of it”.*

The gain from shrinkage. Substituting λ^* directly:

$$\text{MSE}(\lambda^*) = \frac{\sigma^2 \beta_j^2}{\sigma^2 + n \beta_j^2}, \quad \frac{\text{MSE}(\lambda^*)}{\text{MSE}(0)} = \frac{n \beta_j^2}{\sigma^2 + n \beta_j^2} < 1 \quad (\beta_j \neq 0).$$

The ratio tends to 0 as $|\beta_j| \rightarrow 0$: the weaker the signal, the closer the gain of optimal shrinkage gets to 100%; and it tends to 1 as $|\beta_j| \rightarrow \infty$ — **for strong signals the optimal amount of shrinkage tends to zero ($\lambda^* \rightarrow 0$), and shrinkage automatically degenerates into no shrinkage.** This is the quantitative form, under orthogonal design, of “shrinkage never hurts a strong signal and always helps a weak one.” $\square$

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